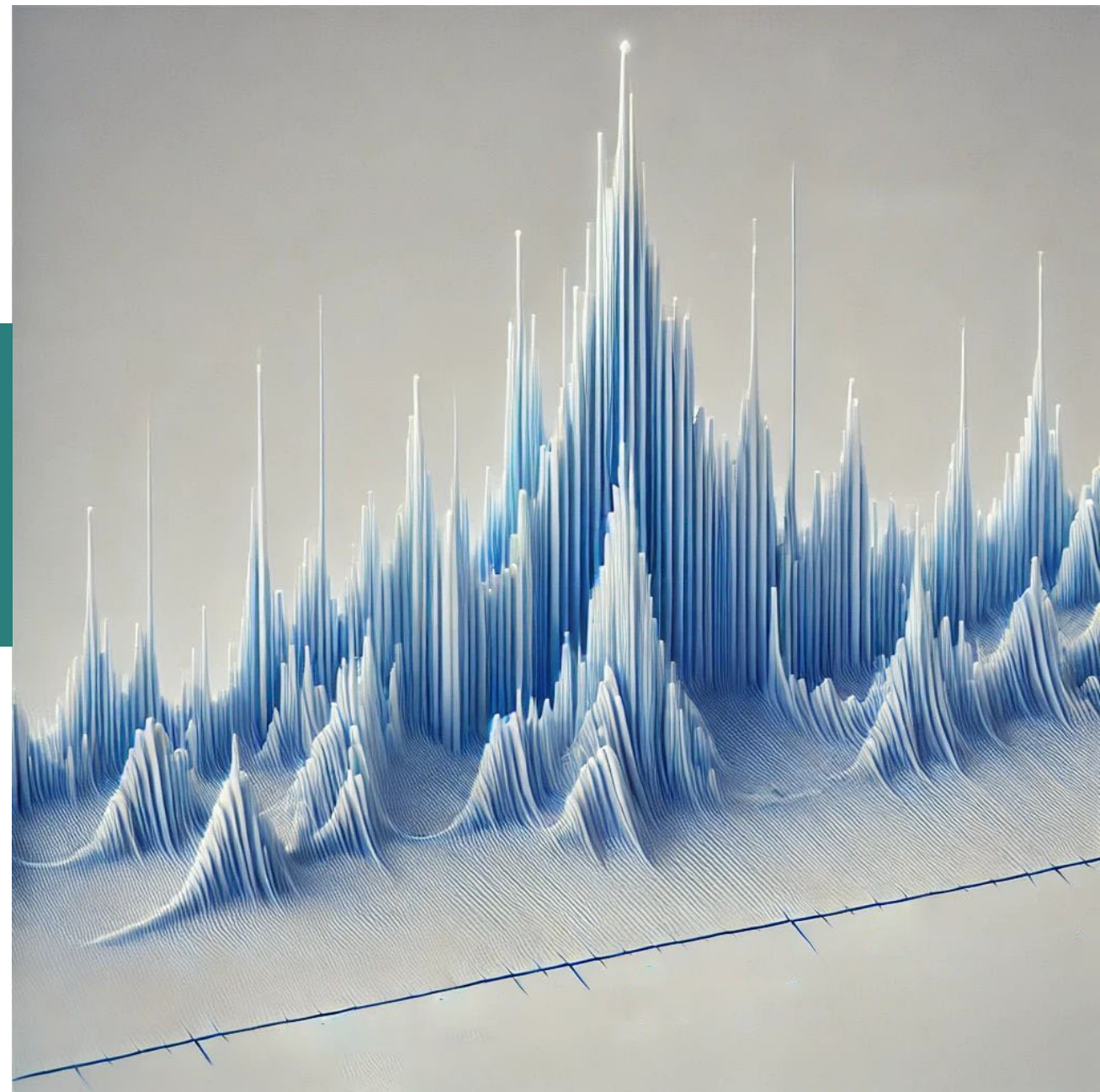


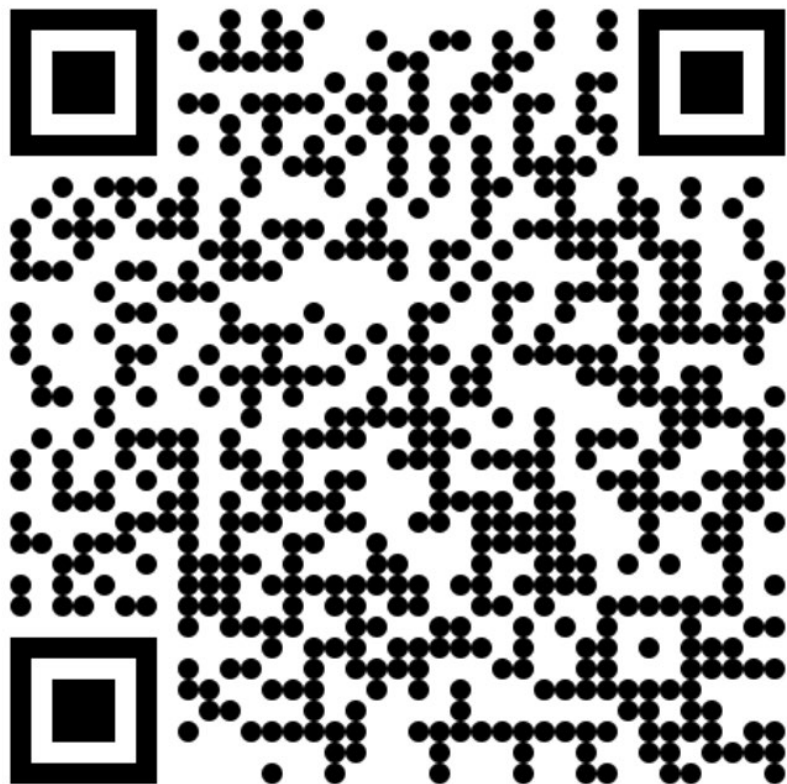
Rietveld Refinement of Powder Diffraction Data

ACerS Workshop: Tools for Visualizing and Understanding the Structure of Crystalline Ceramics

Taylor Sparks
Professor of Materials Science & Engineering, University of Utah



Download the software and tutorial materials now, the files are large



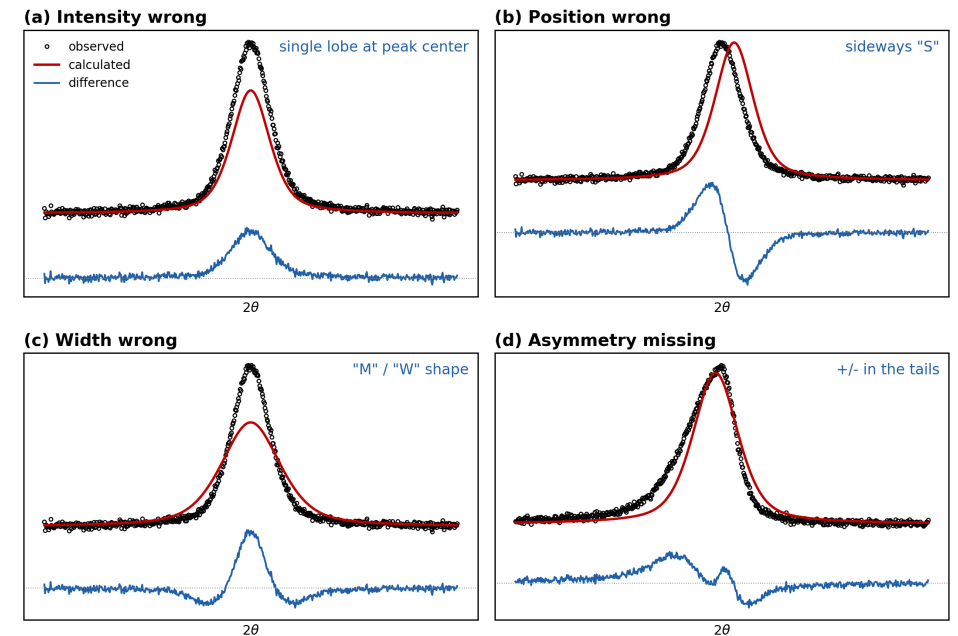
GSAS-II software
~400 MB download, 2 GB installed



Workshop tutorial materials
~30 MB

One technique answers many quantitative questions

- What is Rietveld refinement and how does it work?
- Quantitative analysis of multiphase mixtures
 - Amorphous content
- Atomic positions and site occupancy
- Thermal displacement parameters
- Preferred orientation
- Crystallite size and strain



Rietveld refinement least-squares fits a calculated pattern to your data

- **Hugo Rietveld (1969)**
- Calculated pattern vs measured pattern
- Variable model parameters, adjusted by least squares
- Accounts for measurement errors (shift, zero offset)
- Different from Le Bail or Pawley (profile) fitting: those fit intensities without a structure
- Ab initio structure determination is a separate, harder problem



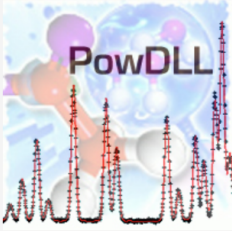
GSAS-II and PowDLL cover the whole workflow for free

PowDLL .NET converter for XRPD files

Home (Lab)DownloadHints'n'TipsScreenshotsFor programmersUpdatesContact

PowDLL is a .NET dynamic link library used for the interconversion procedure between variable formats of Powder X-Ray files. The DLL is capable of handling the most common file formats (binary and ASCII). The library can be used as a reusable component with any .NET language or as a **standalone utility**.

PowDLL can run on windows OS as long as they have dotnet runtimes version 2 (or later) installed (i.e., Windows Vista SP1 or later). Linux is also supported through Wine.



Imports

Bruker/Siemens RAW (versions 1,2,4), Bruker BRML, STOE RAW (plus multi-range files), Scintag RAW (plus multi-range files), Rigaku RAW, Shimadzu RAW, Philips RD, Philips SD, Scintag RD, Panalytical XRDML, INEL Binary, INEL ASCII, Scintag ARD, powderCIF, Sietronics CPI, Riet7 DAT, DBWS, GSAS (CW STD), Jade MDI, Rigaku RIG, Philips UDF, UXD, XDA, XDD, CCDC Mercury XYE, XPOWDER PLV (old and new format), UDF (NEX), ProtoXRD and ASCII XY Files.

Exports

Bruker/Siemens RAW (versions 1,2), Philips RD, Scintag ARD, Sietronics CPI, Riet7 DAT, DBWS, GSAS (CW STD), Jade MDI, Rigaku RIG, Philips UDF, UXD, XDA, XDD, Panalytical XRDML, ASCII XY Files, MS-Excel Multiple XY, Xpowder PLV files.

Citation

Please cite if you find PowDLL useful:

PowDLL, a reusable .NET component for interconverting powder diffraction data: Recent developments, N. Kourkoumelis, ICDD Annual Spring Meetings (ed. Lisa O'Neill), Powder Diffraction, 28 (2013) 137-48.

GSAS-II Home Page



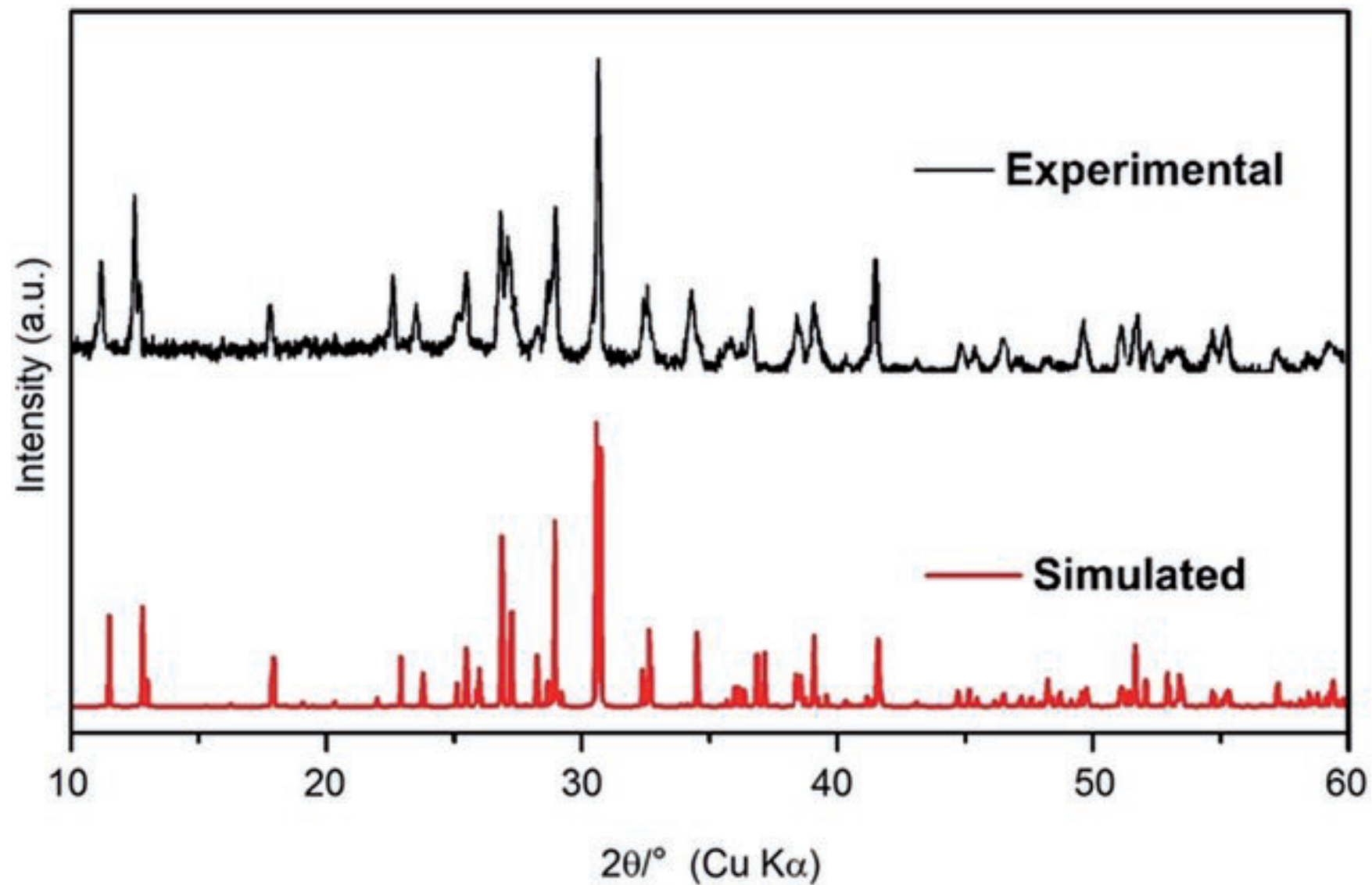
Contents

1. GSAS-II Installation Options
2. Available Documentation for GSAS-II
3. Mailing List
4. Reporting GSAS-II Bugs
5. Information for GSAS-II Developers
6. Compiling GSAS-II
7. Customized Python Installation
8. Web Proxies

Welcome to the home page for GSAS-II, a unique and comprehensive open source Python project for determination of crystal structures and diffraction-based materials characterization for crystalline solids on all scales, from perovskites through proteins, using both powder and single-crystal diffraction and with both x-ray and neutron probes. Refinements can combine measurements from laboratory and synchrotron x-rays, as well as constant wavelength or time-of-flight neutron sources. It provides structure solution and refinement, as well as extensive visualization capabilities.

GSAS-II is made available for free use ([see license](#)) with open access to the [source code](#).

Modern software simulates powder patterns remarkably well

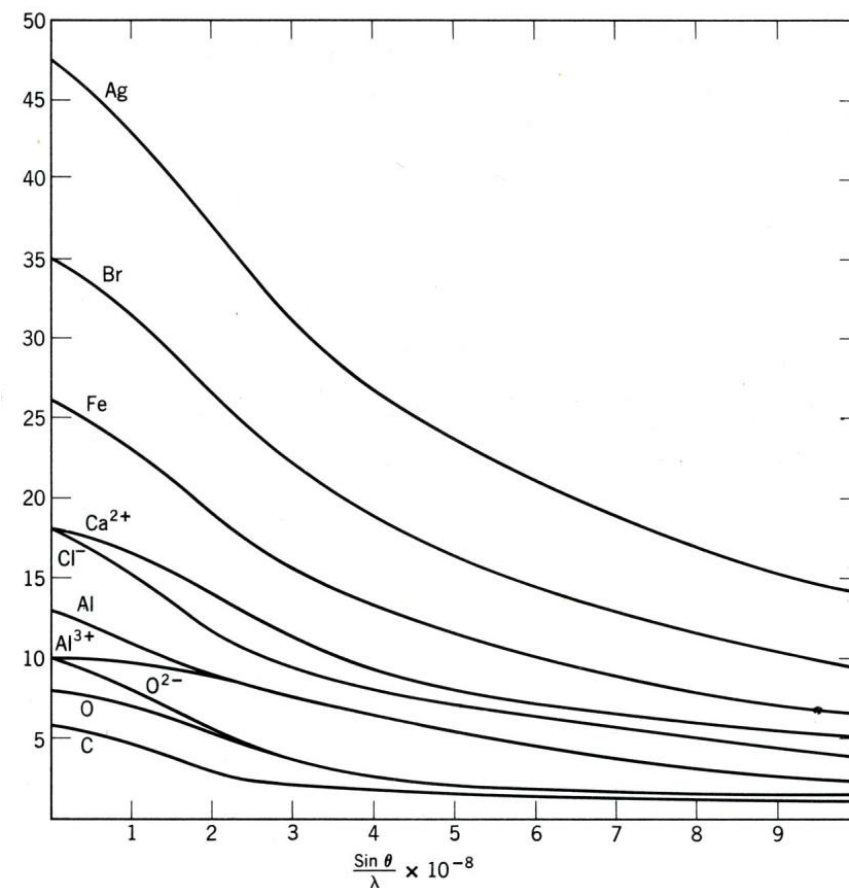


Peak intensity folds in source, sample, and structure factors

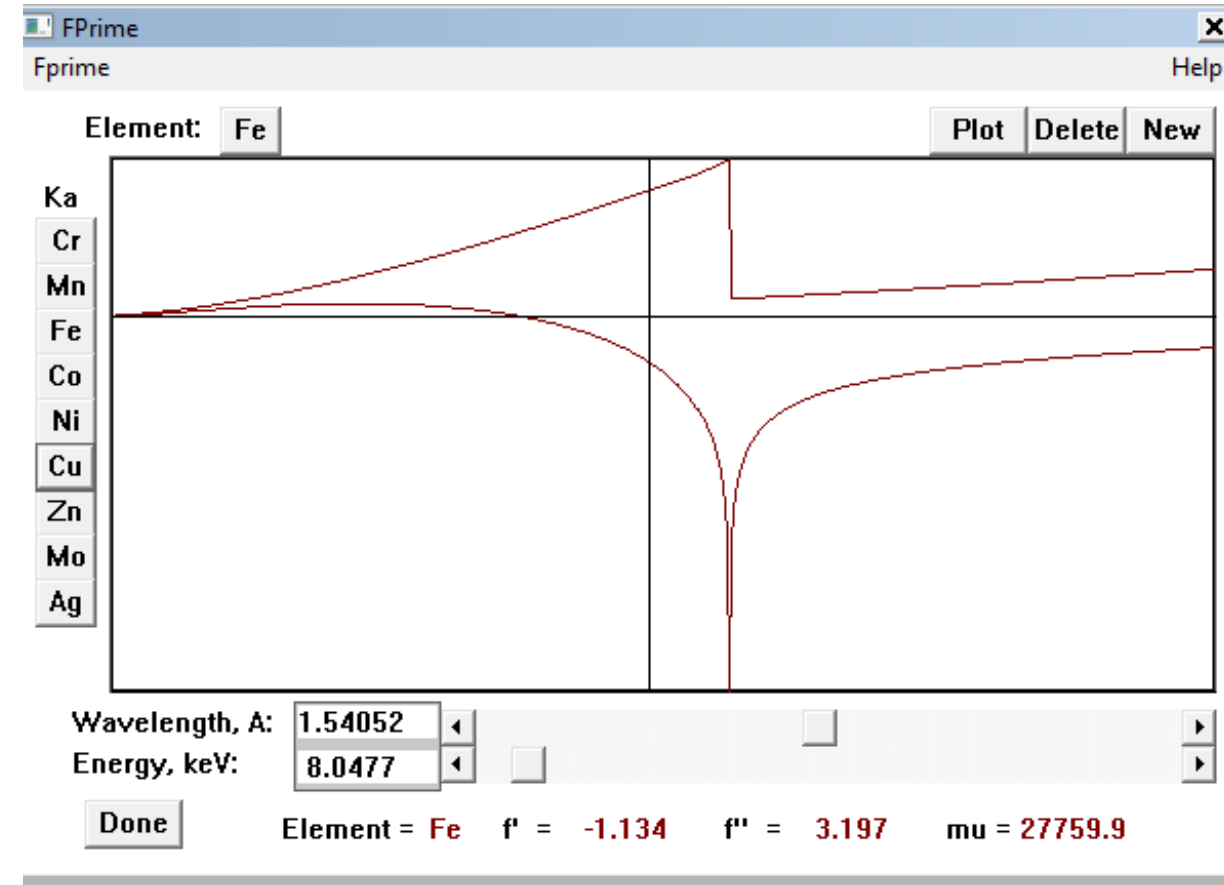
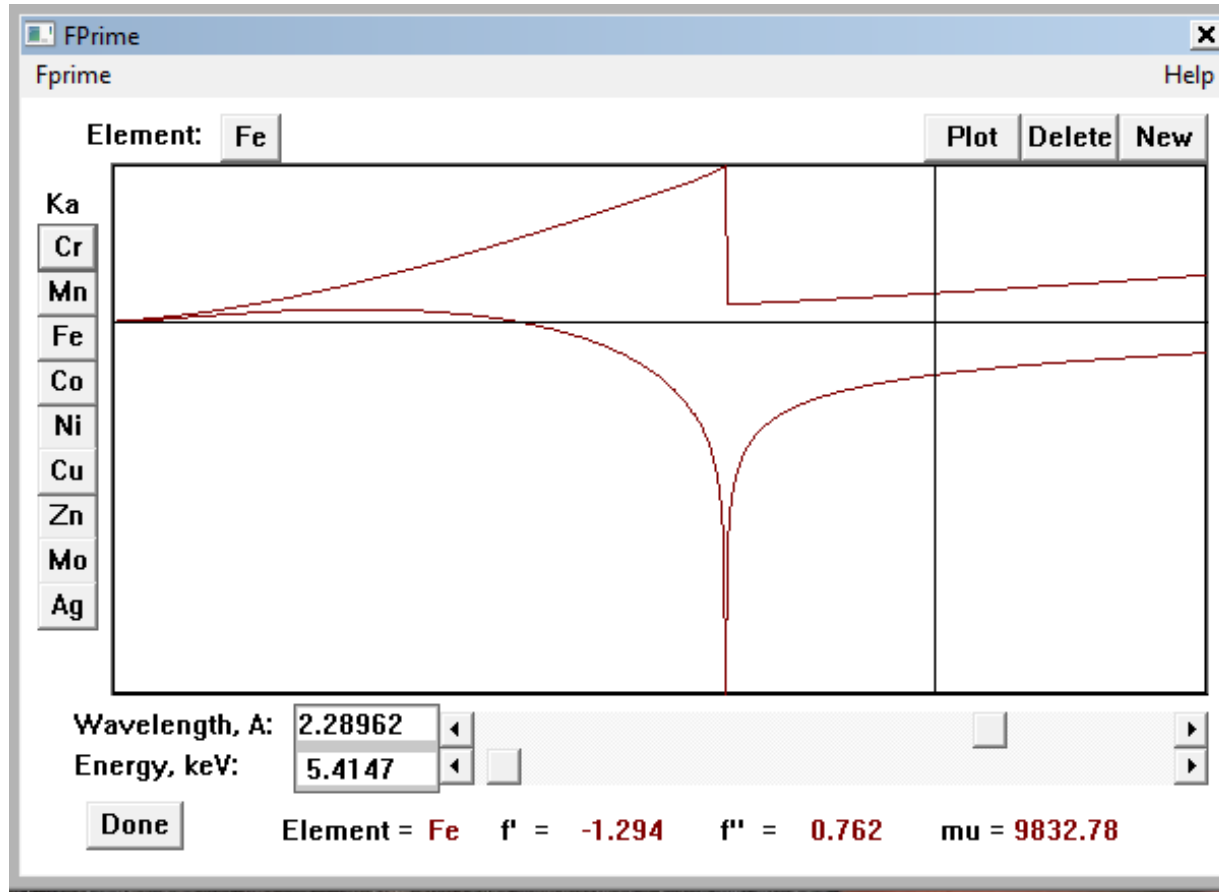
$$I_{(hkl)\alpha} = \frac{I_0 \lambda^3}{64 \pi r} \left(\frac{e^2}{m_e c^2} \right)^2 \frac{M_{(hkl)}}{V_\alpha^2} |F_{(hkl)\alpha}|^2 \left(\frac{1 + \cos^2(2\theta) \cos^2(2\theta_m)}{\sin^2 \theta \cos \theta} \right) \frac{v_\alpha}{\mu_s}$$

$$F_{hkl} = \sum_{j=1}^m N_j f_j \exp[2\pi i(hx_j + ky_j + lz_j)]$$

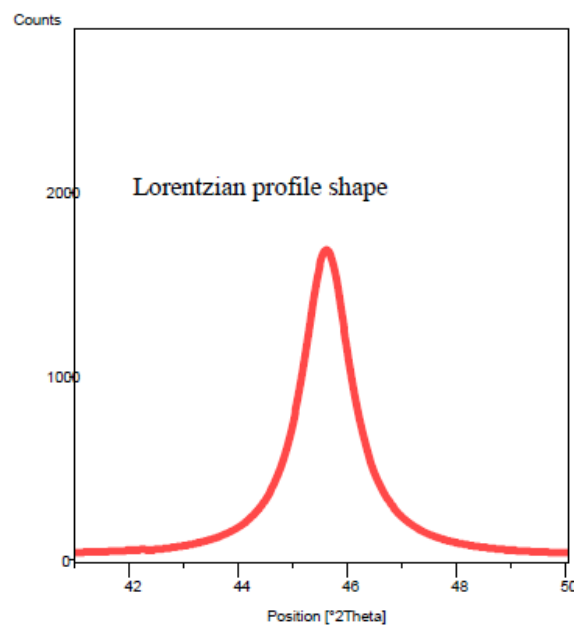
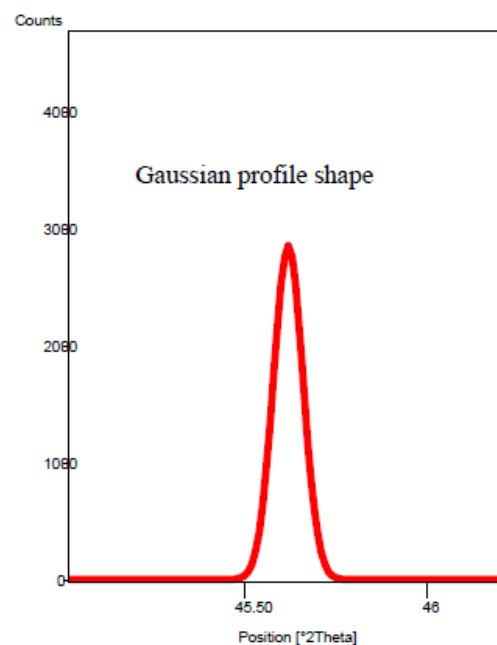
$$|f|^2 = \left(f_0 \exp\left[\frac{-B \sin^2 \theta}{\lambda^2} \right] + \Delta f' \right)^2 + (\Delta f'')^2$$



Radiation choice sets the anomalous scattering factors f' and f''



The instrument dictates Gaussian vs Lorentzian peak character



$$H_k = \left(U \tan^2 \theta + V \tan \theta + W \right)^{1/2}$$



Every parameter changes the pattern in one of three ways

WHERE peaks sit

Lattice parameters (a, b, c, angles)

Zero offset

Sample displacement

WHAT SHAPE they have

Gaussian widths: U, V, W

Lorentzian widths: X, Y

Asymmetry (axial divergence)

Size and strain broadening

HOW TALL they are

Scale factor(s)

Atomic positions (x, y, z)

Site occupancies

Thermal parameters (U_{iso} , U_{ij})

Preferred orientation

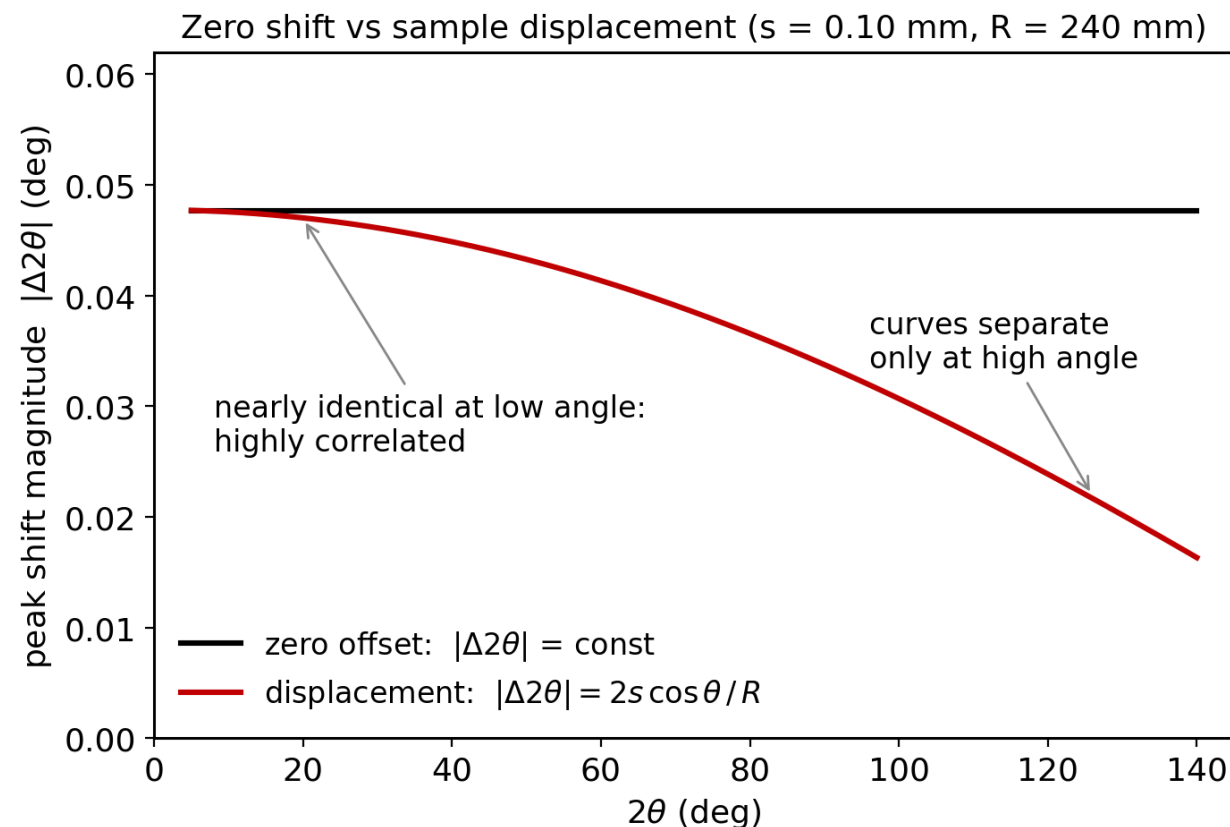
Absorption, roughness

Diagnosis strategy: when the fit is bad, ask which bucket is wrong. Each bucket leaves a different fingerprint in the difference curve.

Peak positions must match before anything else can refine

- **Two rival corrections:**

- Zero offset: $\Delta 2\theta = \text{const.}$
- Displacement: $\Delta 2\theta = -2s \cos\theta/R$
- Nearly identical at low angle, so the two are highly correlated
- Fix: internal standard (NIST Si 640) pins one of them
- If positions are off, nothing downstream can converge (McCusker et al., 1999)



Peak widths encode real microstructure through U, V, W, X, Y

- **Gaussian part (Caglioti, 1958):**

- $\text{FWHM}^2 = U \tan^2\theta + V \tan\theta + W$

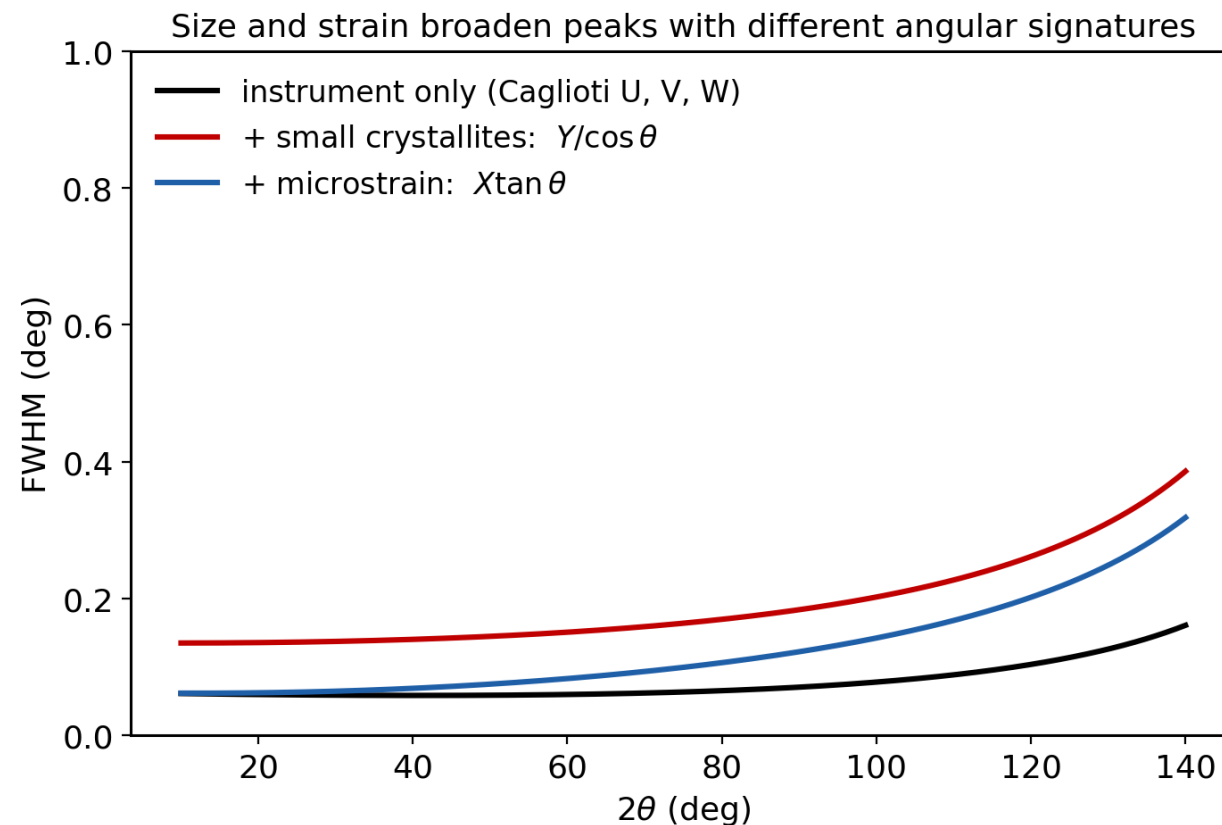
- **Lorentzian part:**

- $\text{FWHM} = X \tan\theta + Y / \cos\theta$

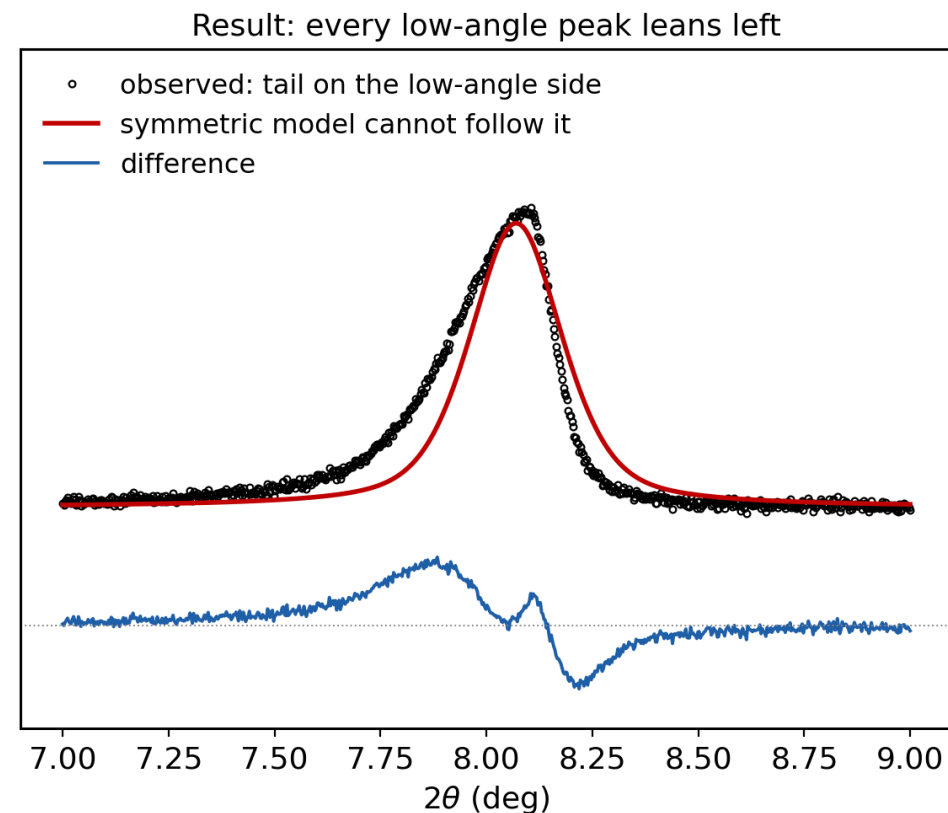
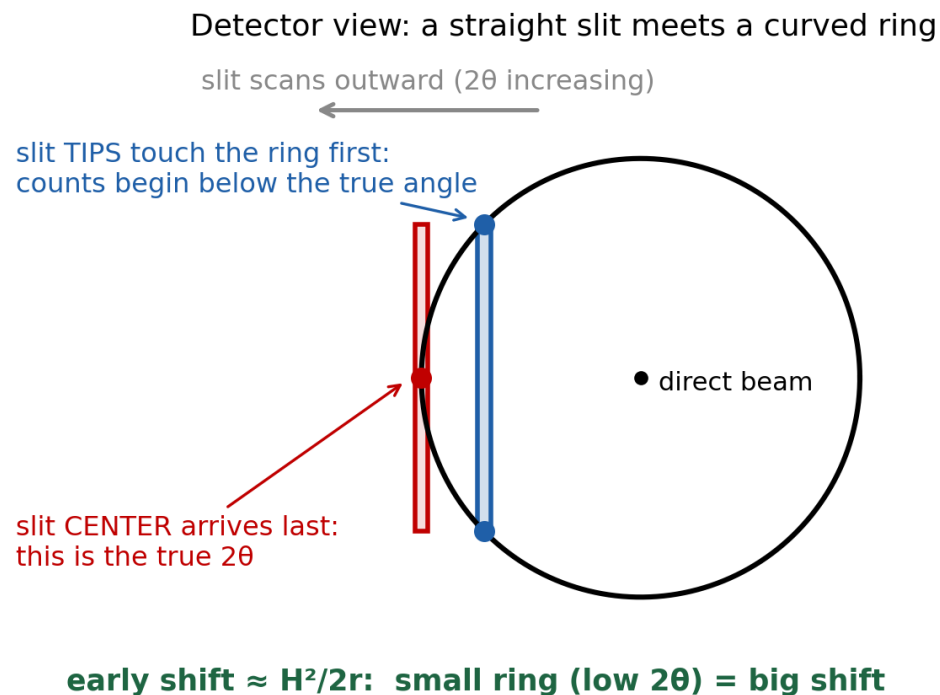
- X grows like strain broadening ($\tan\theta$)

- Y grows like size broadening ($1/\cos\theta$, Scherrer)

- Profile terms are physics, not fudge factors



Axial divergence makes low-angle peaks lean; correct it, don't absorb it

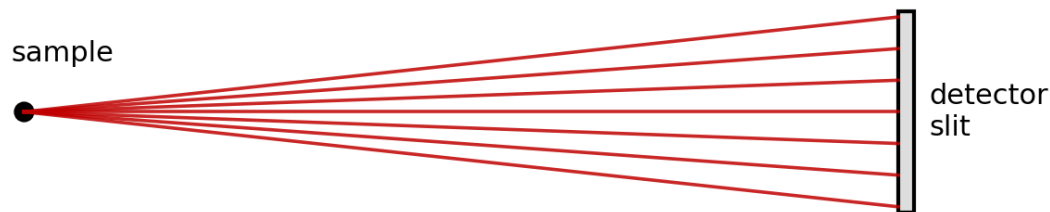


- Diffraction makes cones; on the detector they are rings, and low-angle rings are small and strongly curved
- The receiving slit is straight and tall: scanning outward, its tips clip the ring before its center gets there, so counts arrive early and pile up on the low-angle side
- The early shift scales as $H^2/2r$, which is why the lean is worst at low 2θ (and on high-resolution instruments)
- Model it with the Finger-Cox-Jephcoat correction (built into GSAS-II); never absorb it into peak width

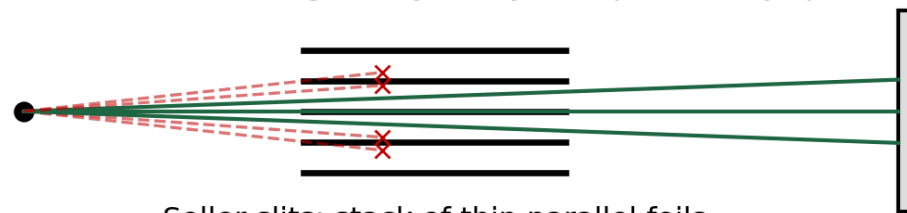
Soller slits tame axial divergence, at the cost of intensity

Side view (axial direction is vertical)

no collimation: rays with large axial divergence all reach the slit

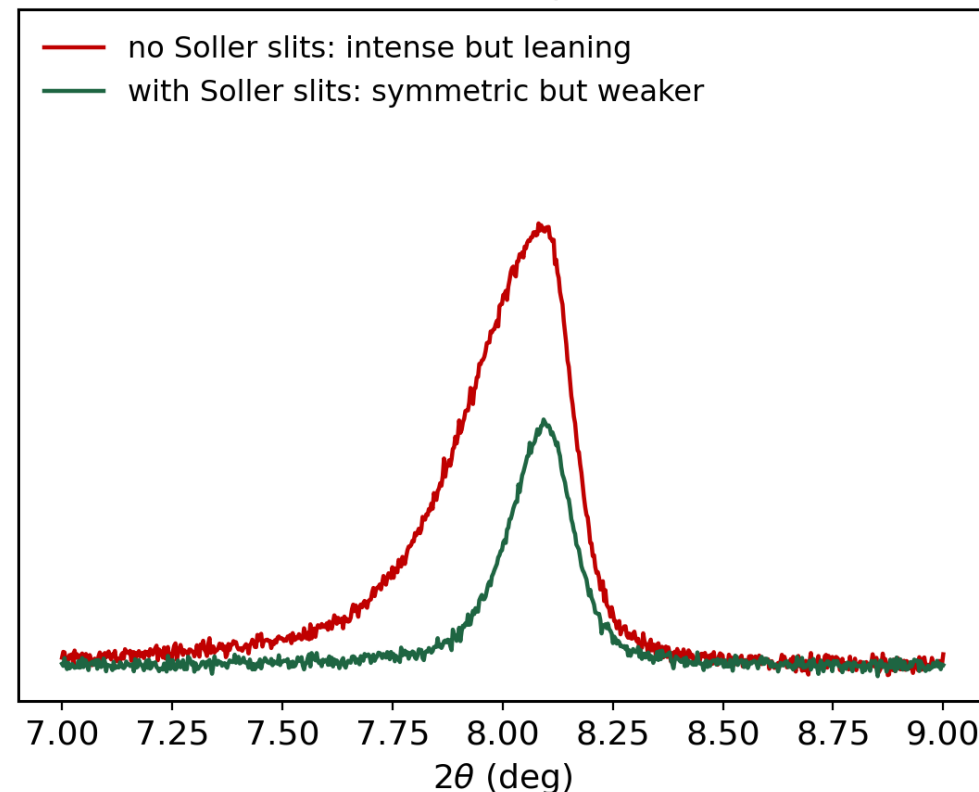


foils absorb divergent rays; only near-parallel rays pass



Soller slits: stack of thin parallel foils

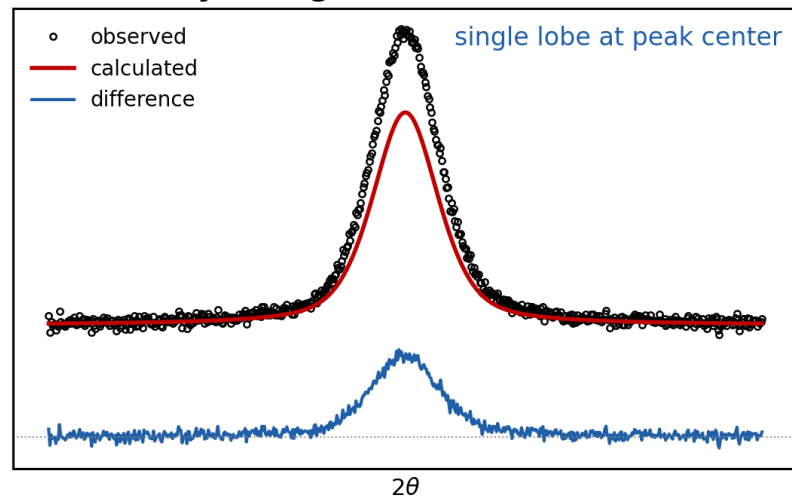
The trade: cleaner shape, fewer counts



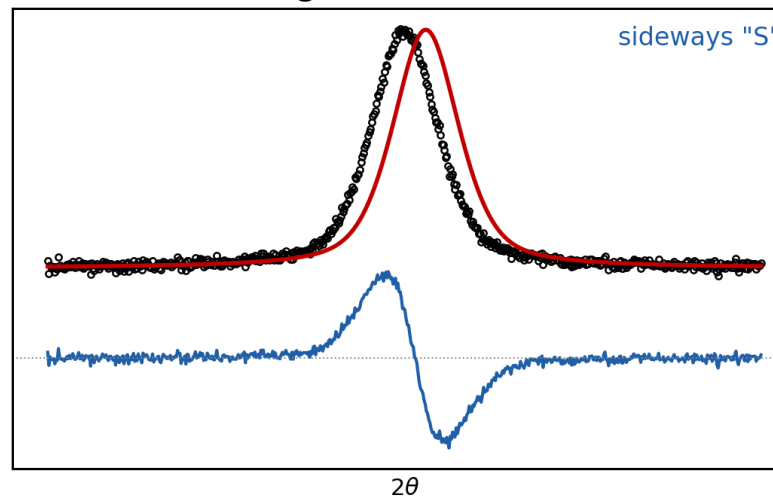
- A stack of thin parallel foils limits out-of-plane divergence to roughly (foil spacing) / (foil length)
- Less axial divergence = shorter low-angle tail and more accurate peak positions
- The price is counts: every blocked ray is simply lost, so counting times grow
- Some asymmetry always survives, so the Finger-Cox-Jephcoat correction still applies

The difference curve tells you which parameter is wrong

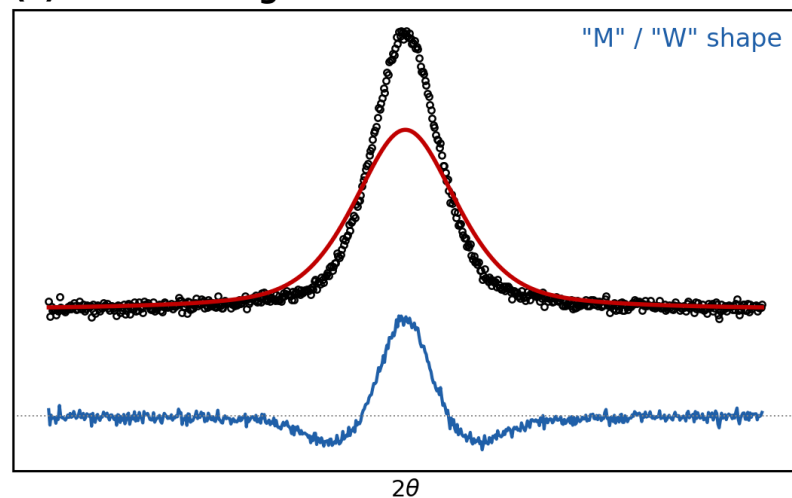
(a) Intensity wrong



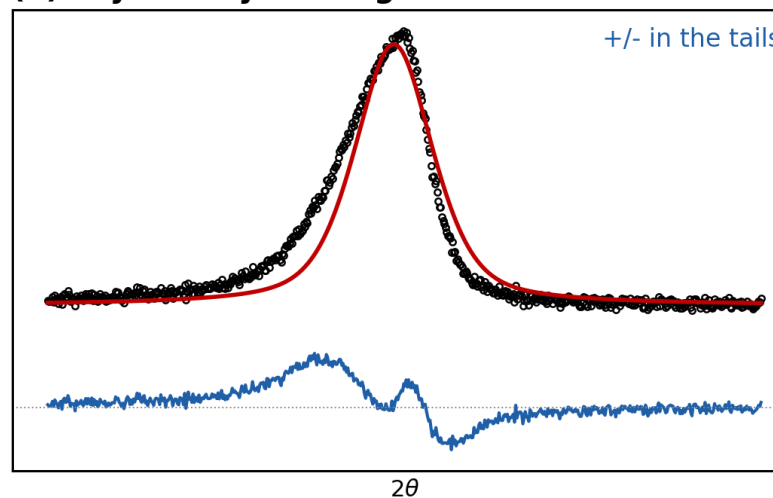
(b) Position wrong



(c) Width wrong



(d) Asymmetry missing



Rp, Rwp, and chi-squared put numbers on the fit

- Rp: plain difference between observed and calculated intensities
- Rwp: weighted, so strong peaks matter more than weak ones; this is what least squares minimizes
- **Common guideline: Rwp < 10% or $\chi^2 < 4$ for a good fit**

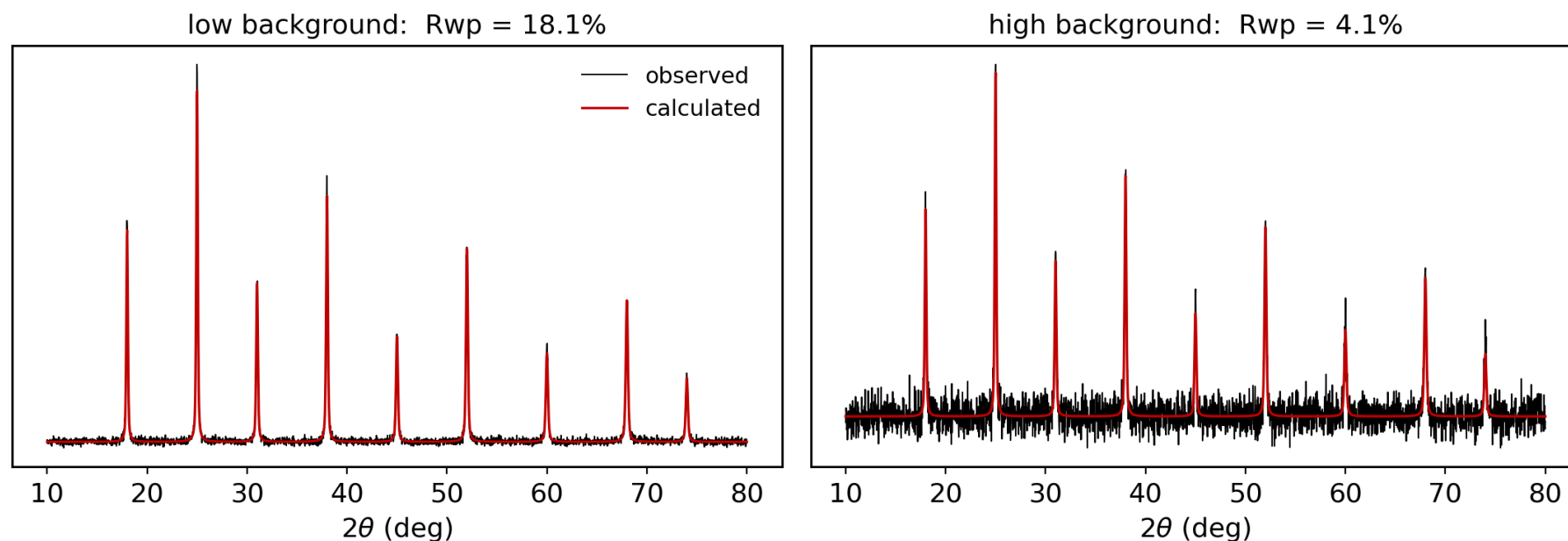
$$R_P = \frac{\sum |y_{io} - y_{ic}|}{\sum y_{io}} \quad R_{wp} = \left[\frac{\sum w_i (y_{io} - y_{ic})^2}{\sum w_i y_{io}^2} \right]^{1/2}$$

$$GOF = \chi^2 = \left[\frac{R_{wp}}{R_{exp}} \right]^2$$

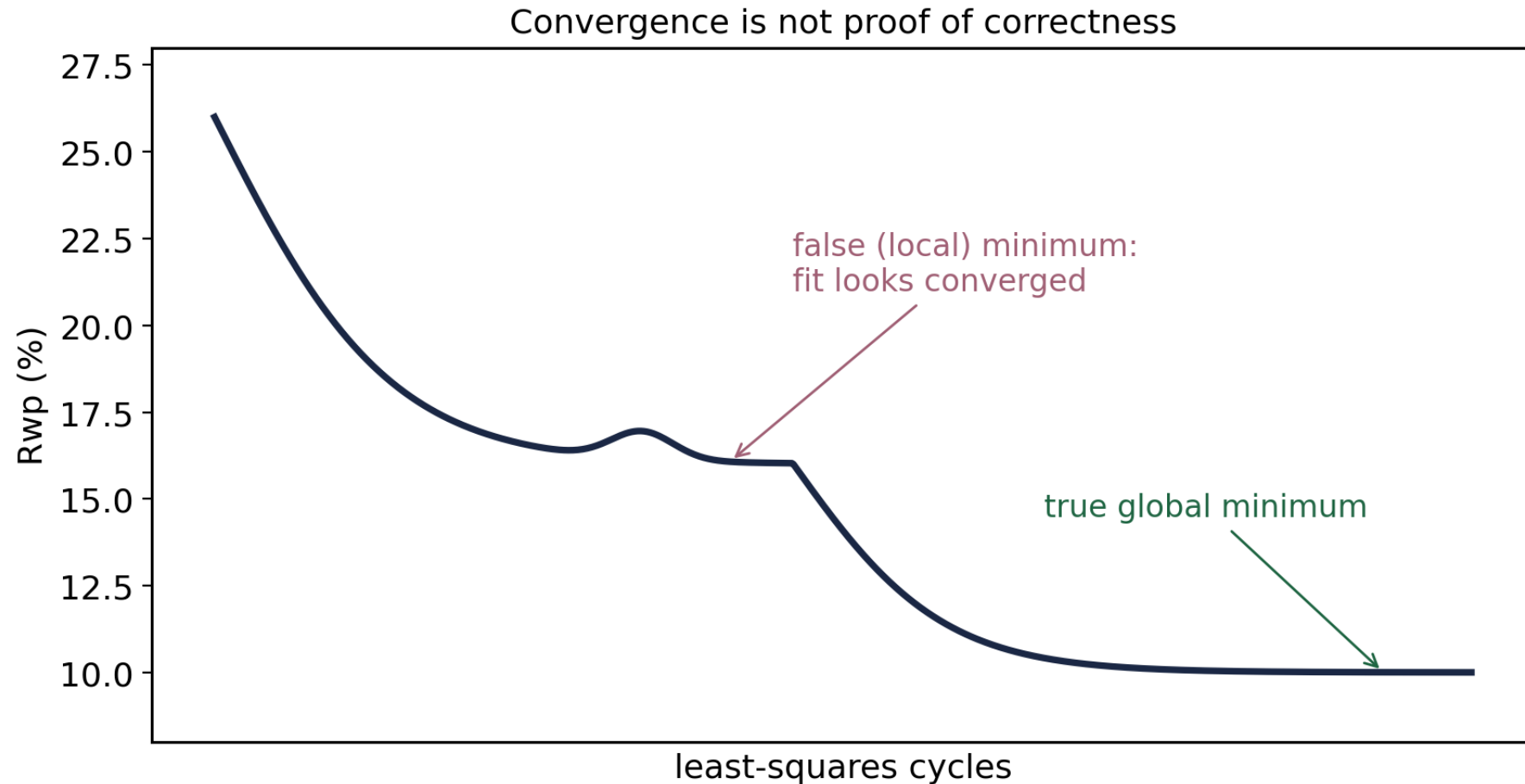
A good Rwp can hide a bad fit, and vice versa

- Background sits in the denominator: high background flatters Rwp for the very same peak misfit
- Never compare Rwp across instruments: TOF neutron at 3% and lab XRD at 10% can be equally good fits
- Your floor is the Le Bail Rwp; judge fits with the plot and chemical sense, not the number (Toby, 2006)

Identical peak misfit, but background sits in the denominator of Rwp



Convergence to a minimum is not proof you found THE minimum

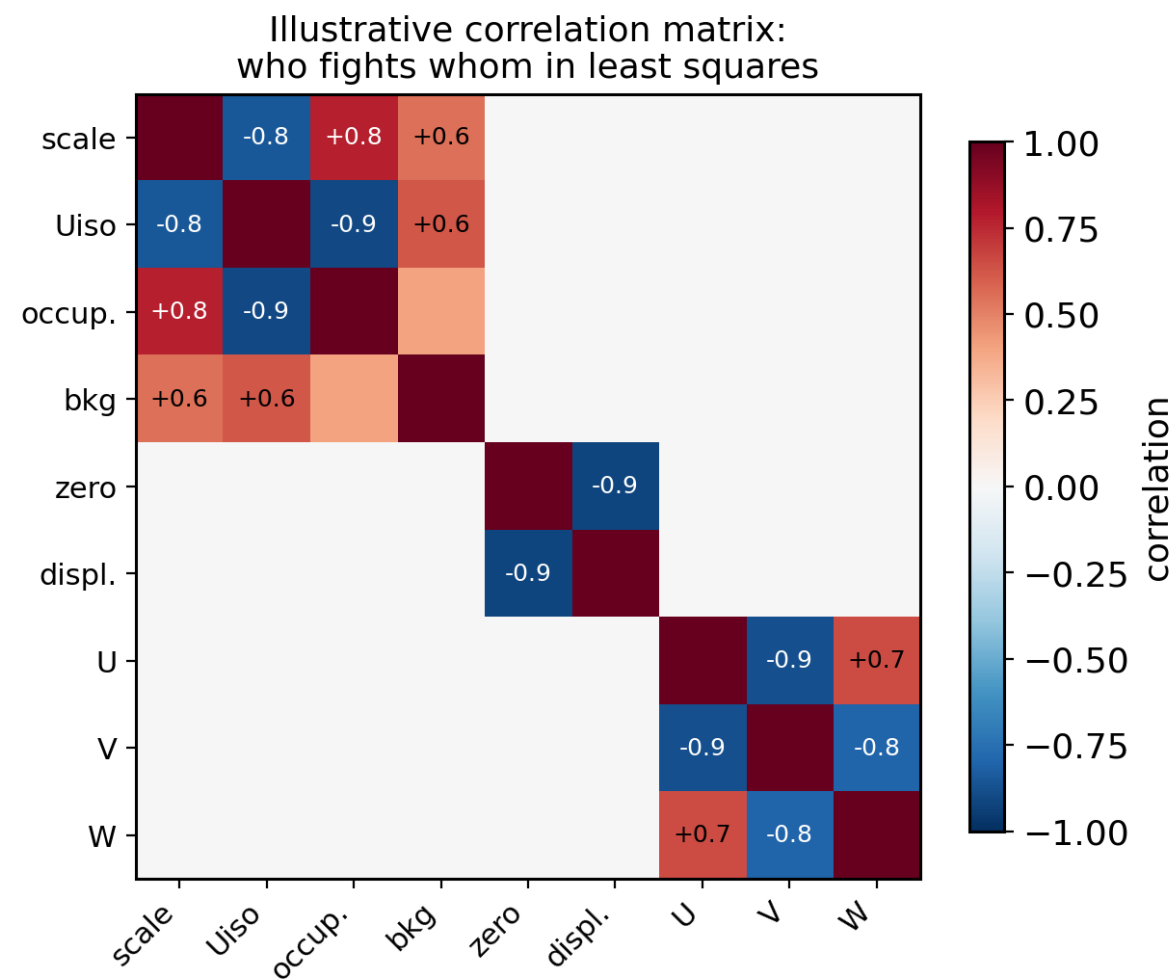


"A Rietveld refinement is never finished, only abandoned." P. W. Stephens

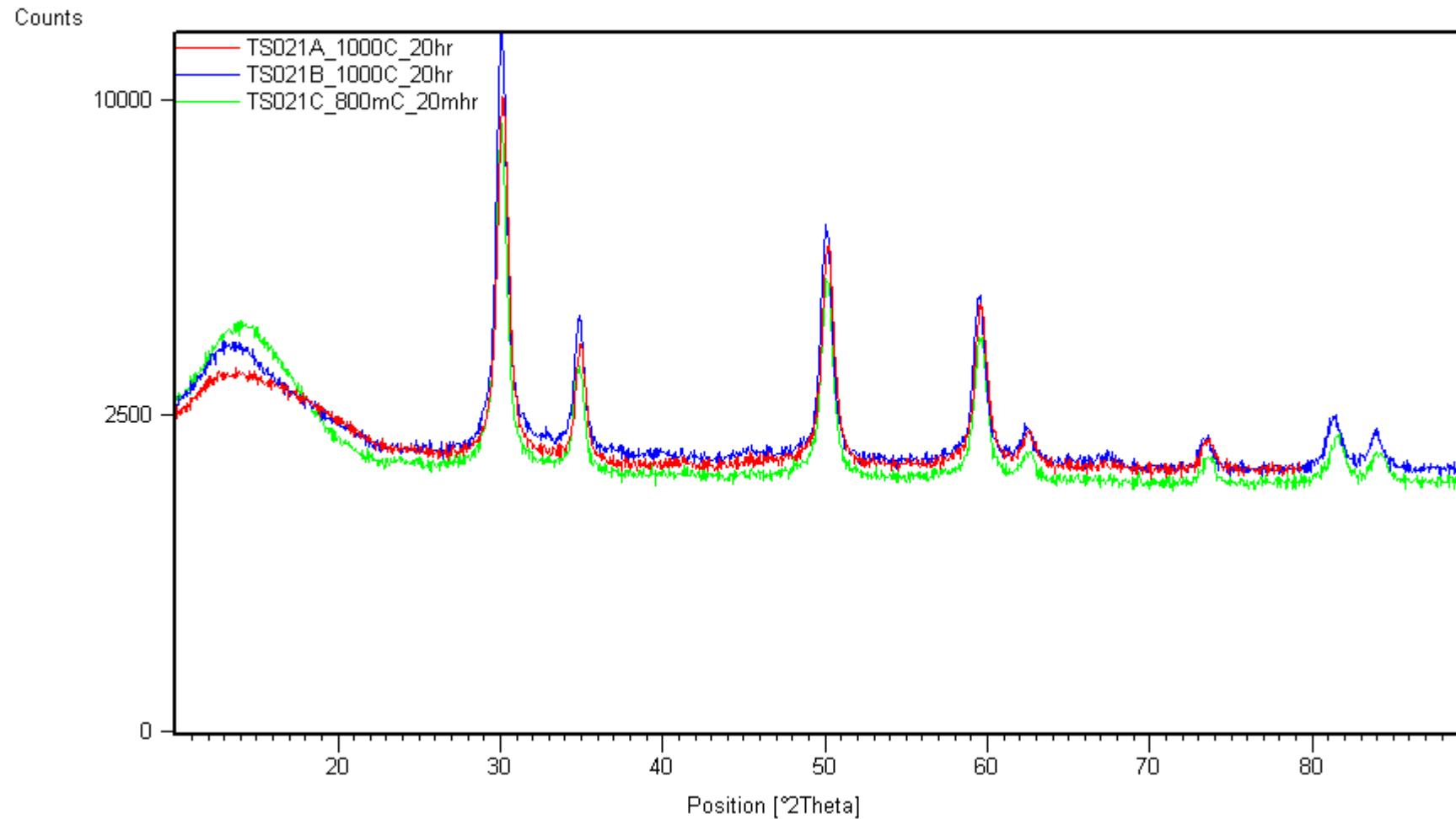
Refinement turn-on order is correlation management

1. Scale + background
2. Lattice + zero (or displacement)
3. Profile: W first, then U, V, X, Y
4. Positions: heavy atoms, then light
5. Uiso (constrain similar atoms early)
6. Occupancies last, then all together

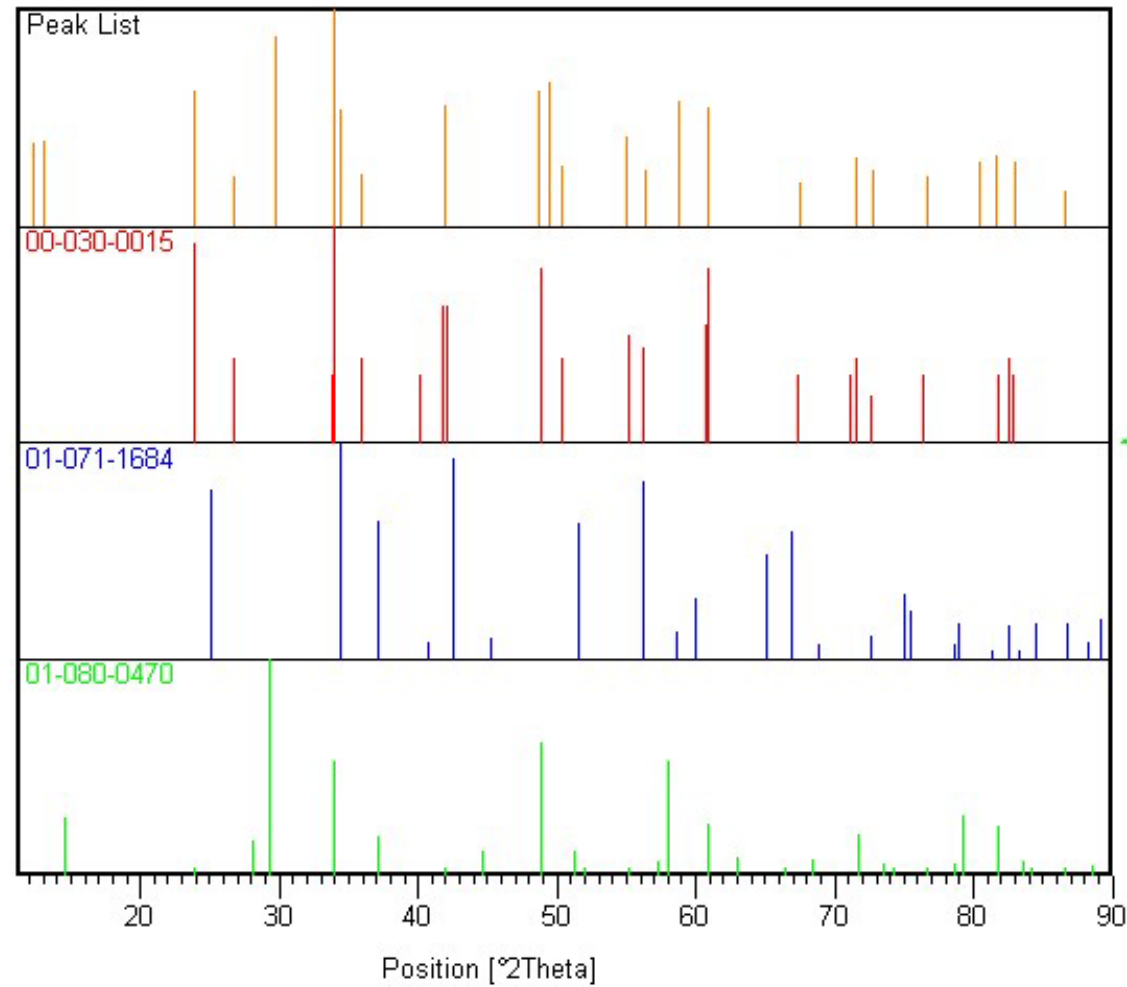
Rule of thumb: observations per parameter ≥ 3 , ideally 5



Multiphase analysis starts by matching the peak list to phases



A three-phase Rietveld fit resolves the mixture directly

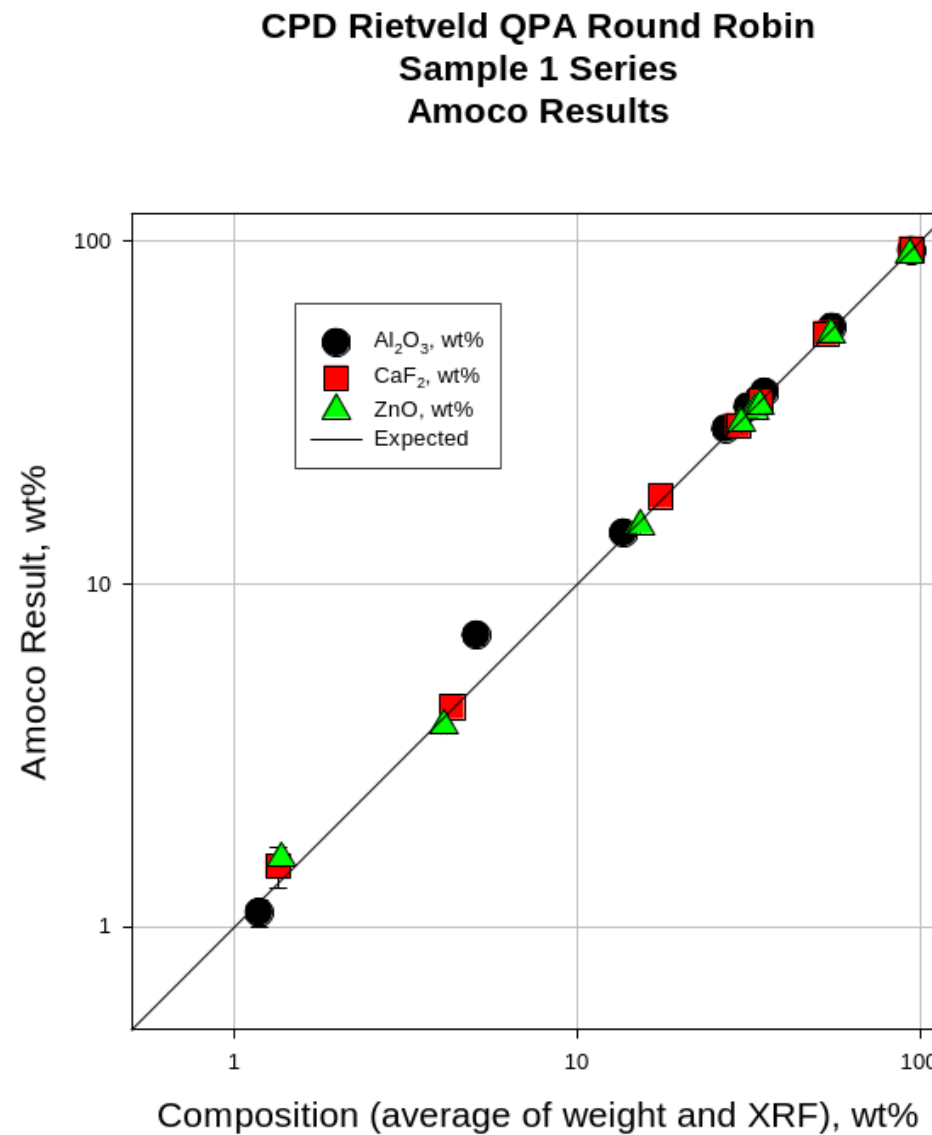


The SMZ relation converts scale factors into weight fractions

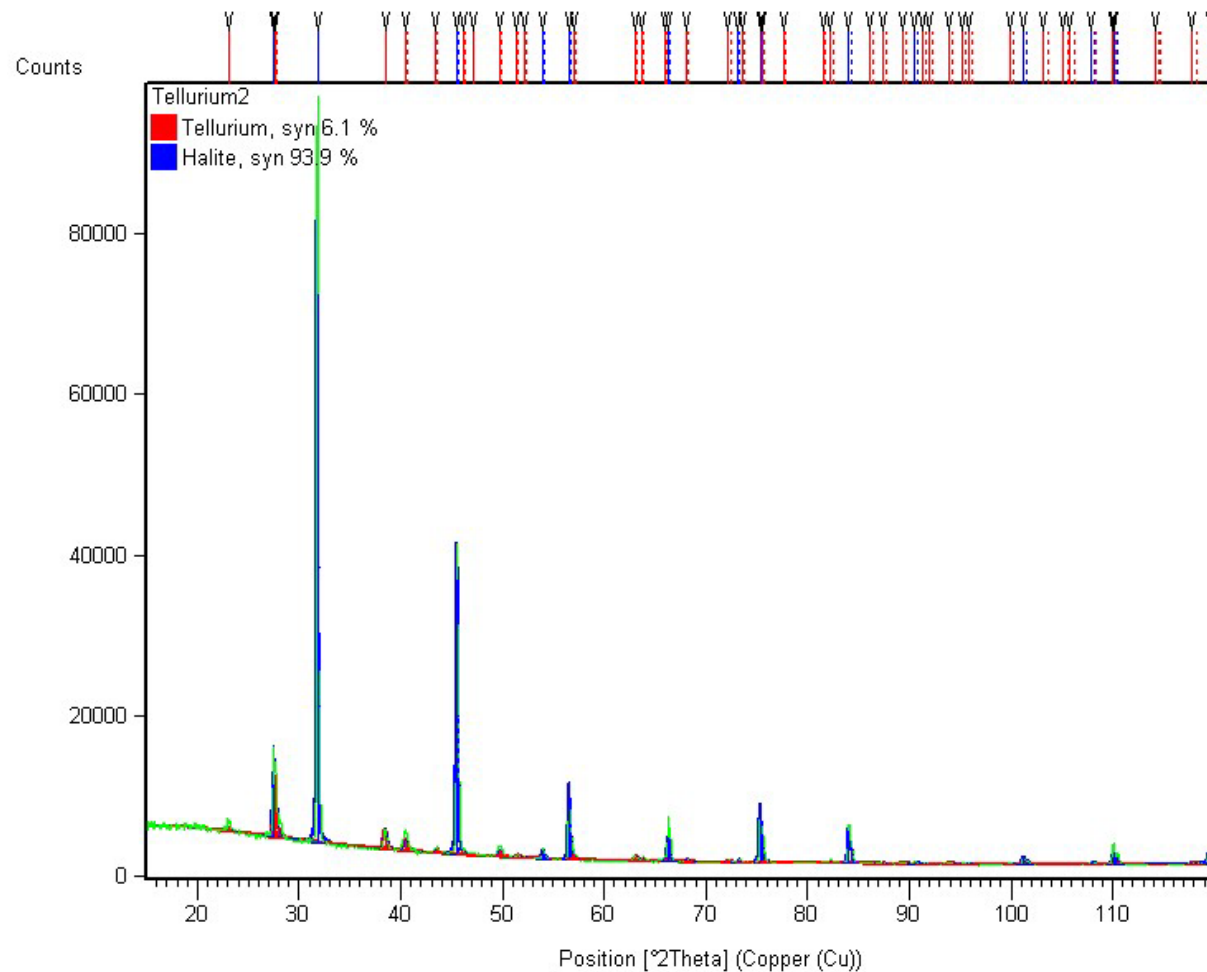
- S is the phase scale factor, proportional to the number of unit cells measured
- M is the molecular weight
- Z is the number of formula units per unit cell
- The product SMZ is proportional to the concentration of each phase

$$X_{\alpha} = \frac{S_{\alpha} M_{\alpha} Z_{\alpha}}{\sum_i S_i M_i Z_i}$$

Round-robin studies show quantitative analysis is accurate to a few percent



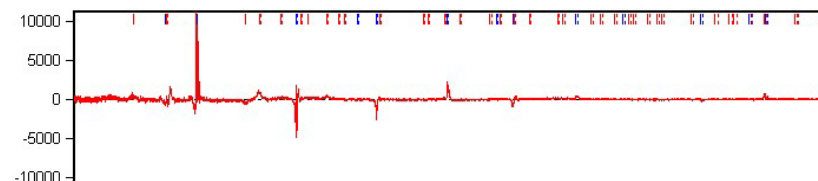
Rietveld quantifies 6% tellurium recovered from a seawater medium



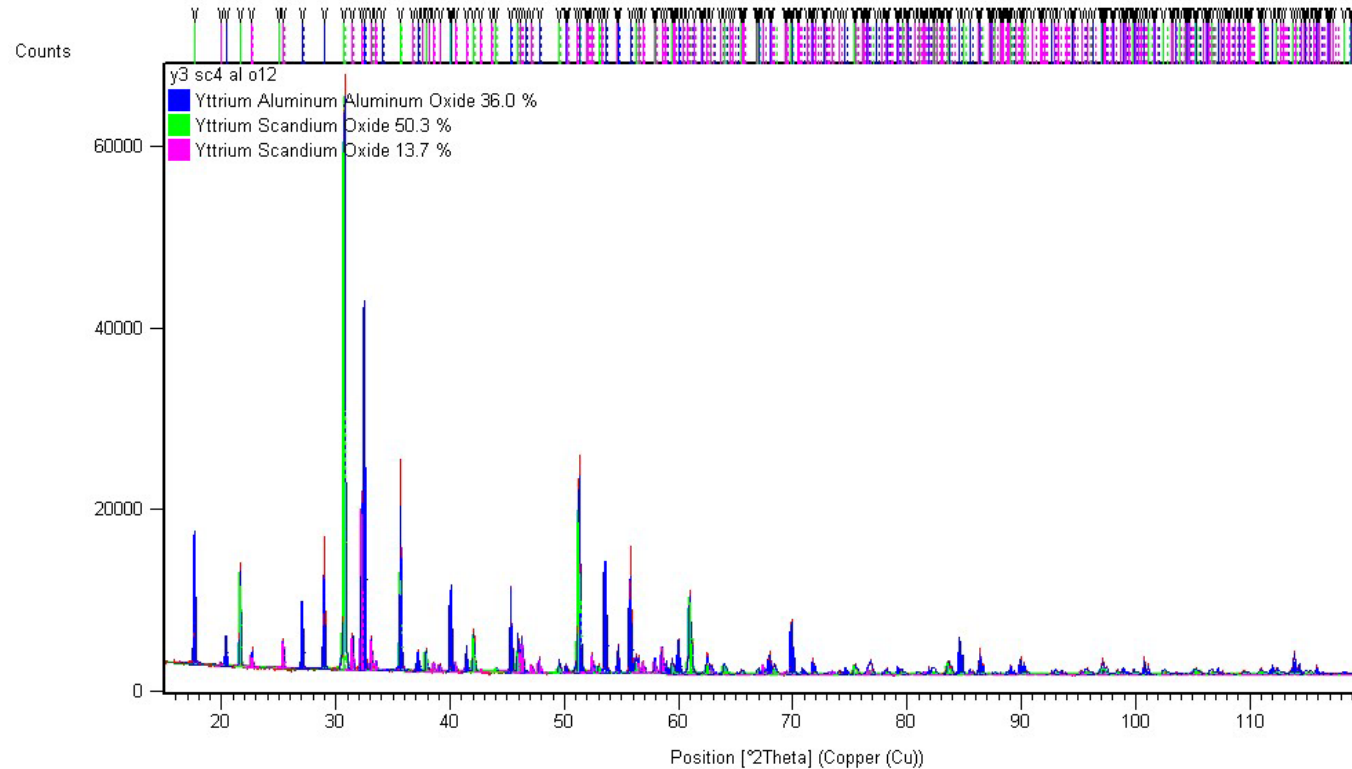
93.9% NaCl

6.1% Te

Rwp = 6.9%



The intended Y3Sc4AlO12 actually crystallized as three phases

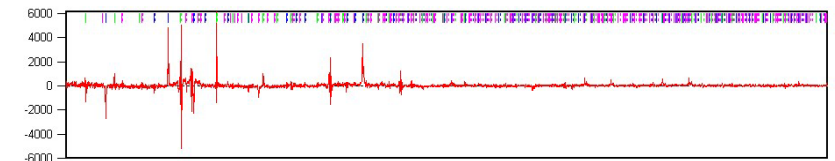


36% YAG (Ia-3d)

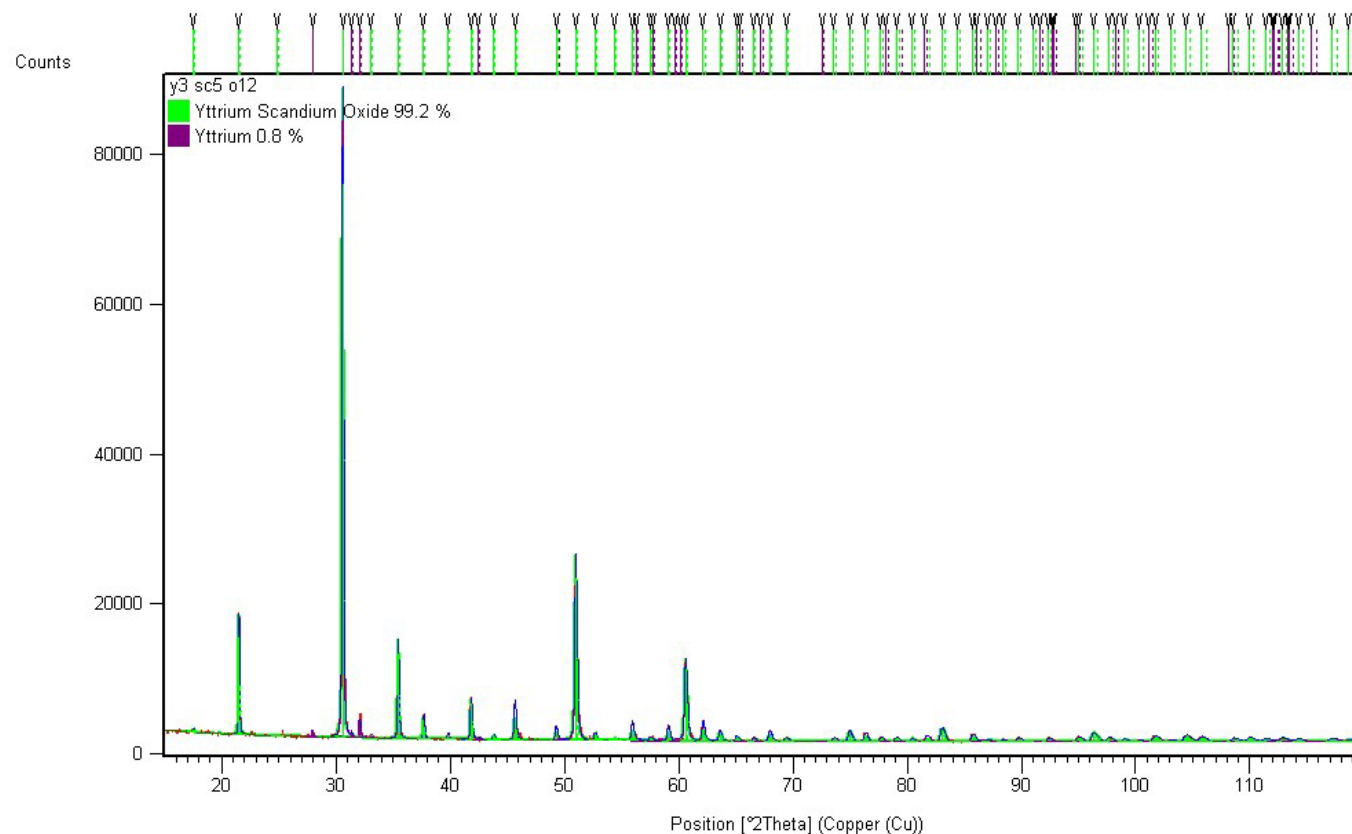
50.3% YScO3 (Ia-3)

13.7% YScO3 (Pnma)

Rwp = 5.3%



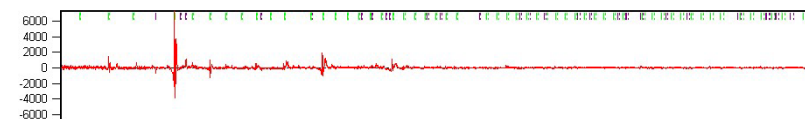
Y3Sc5O12 instead formed nearly phase-pure bixbyite



99.2% Y₃/4Sc₅/4O₃ (Ia-3)

0.8% Y (P63/mmc)

R_wp = 4.7%



The Debye equation extends analysis to amorphous content

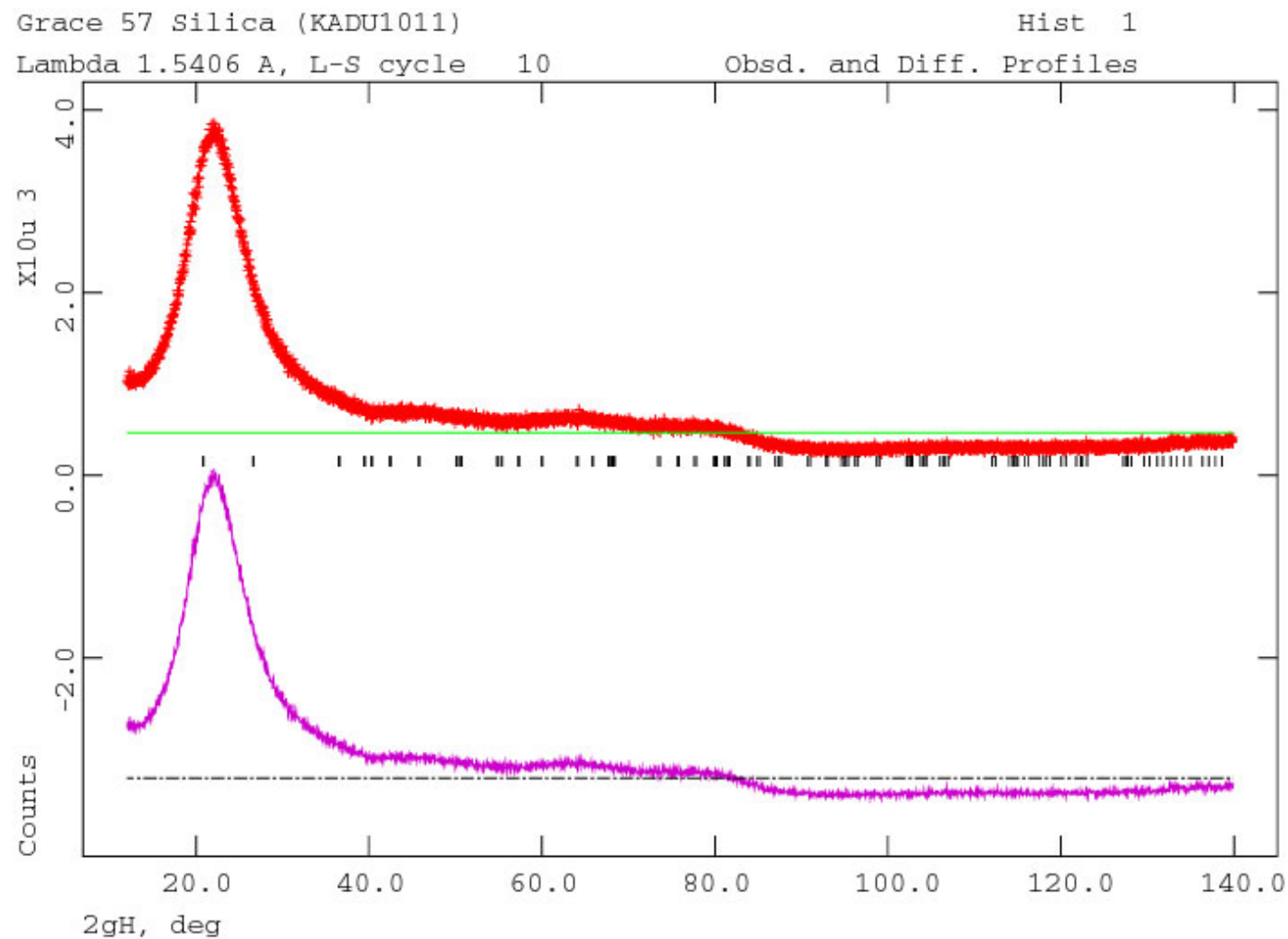
- Percent amorphous material can be determined if a standard is added in a known amount
- Terms describe interatomic distance and vibration amplitude
- A radial distribution function is needed to determine bond distances for analysis

$$I(\theta) = \sum_i f_i^2(\theta) + 2 \sum_i \sum_{j>i} f_i(\theta) f_j(\theta) \frac{\sin(4\pi r_{ij} \sin \theta / \lambda)}{4\pi r_{ij} \sin \theta / \lambda}$$

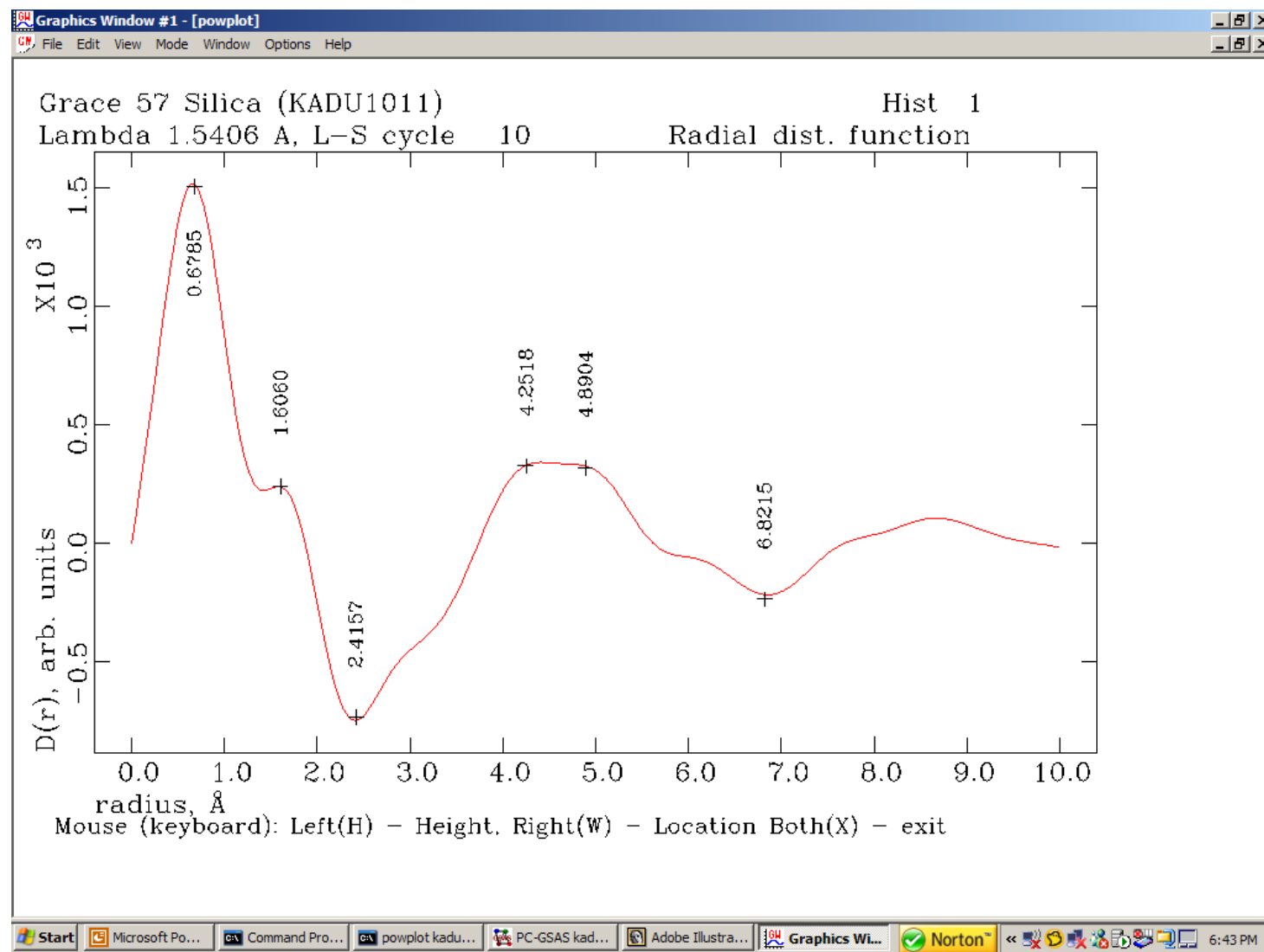
$$A \frac{\sin(QR_i)}{QR_i} \exp\left(-\frac{1}{2} B_i Q^2\right)$$

A: amplitude R_i : distance B_i : vibration

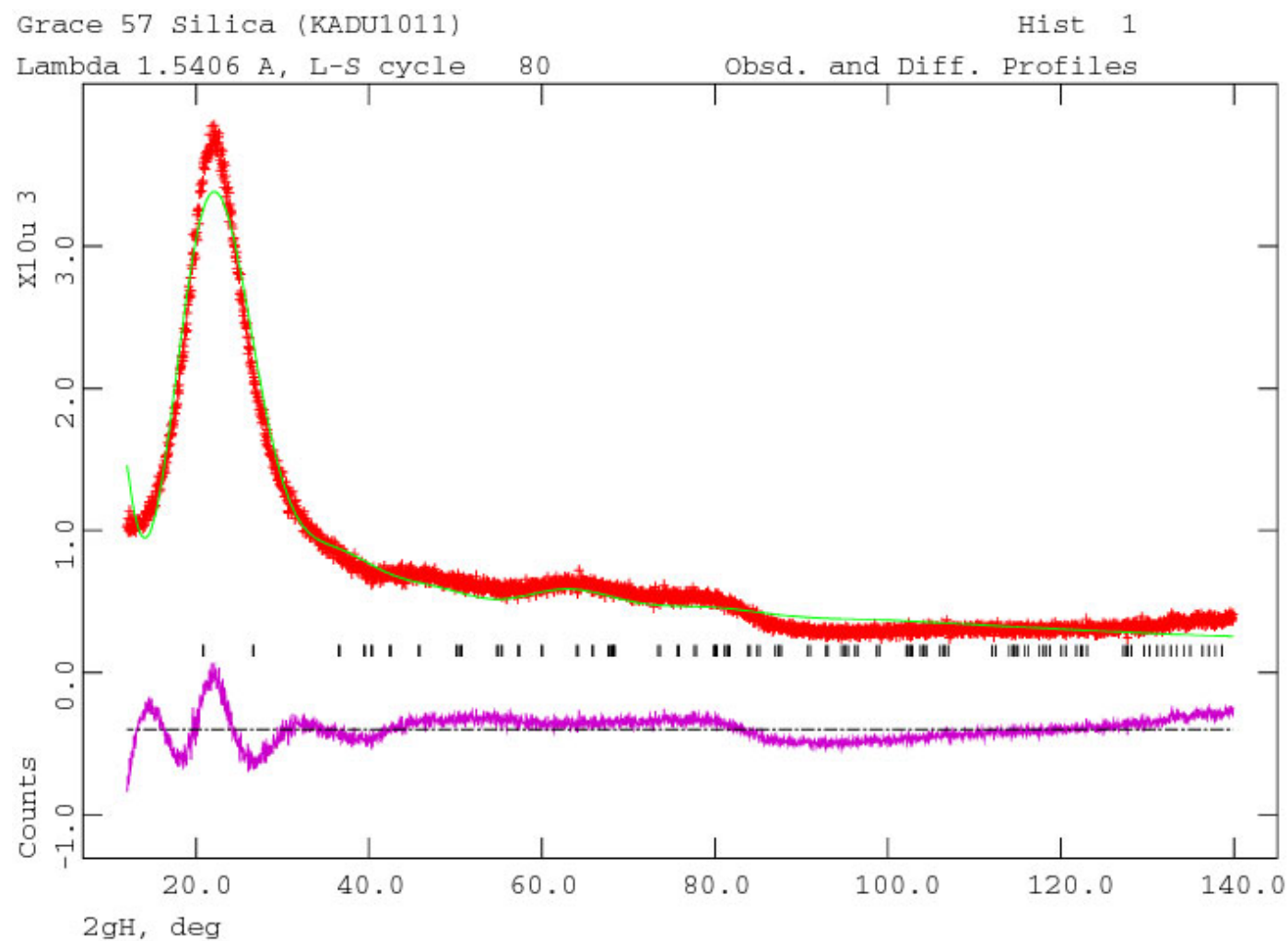
Glassy silica's diffuse scattering can be fit, not just subtracted



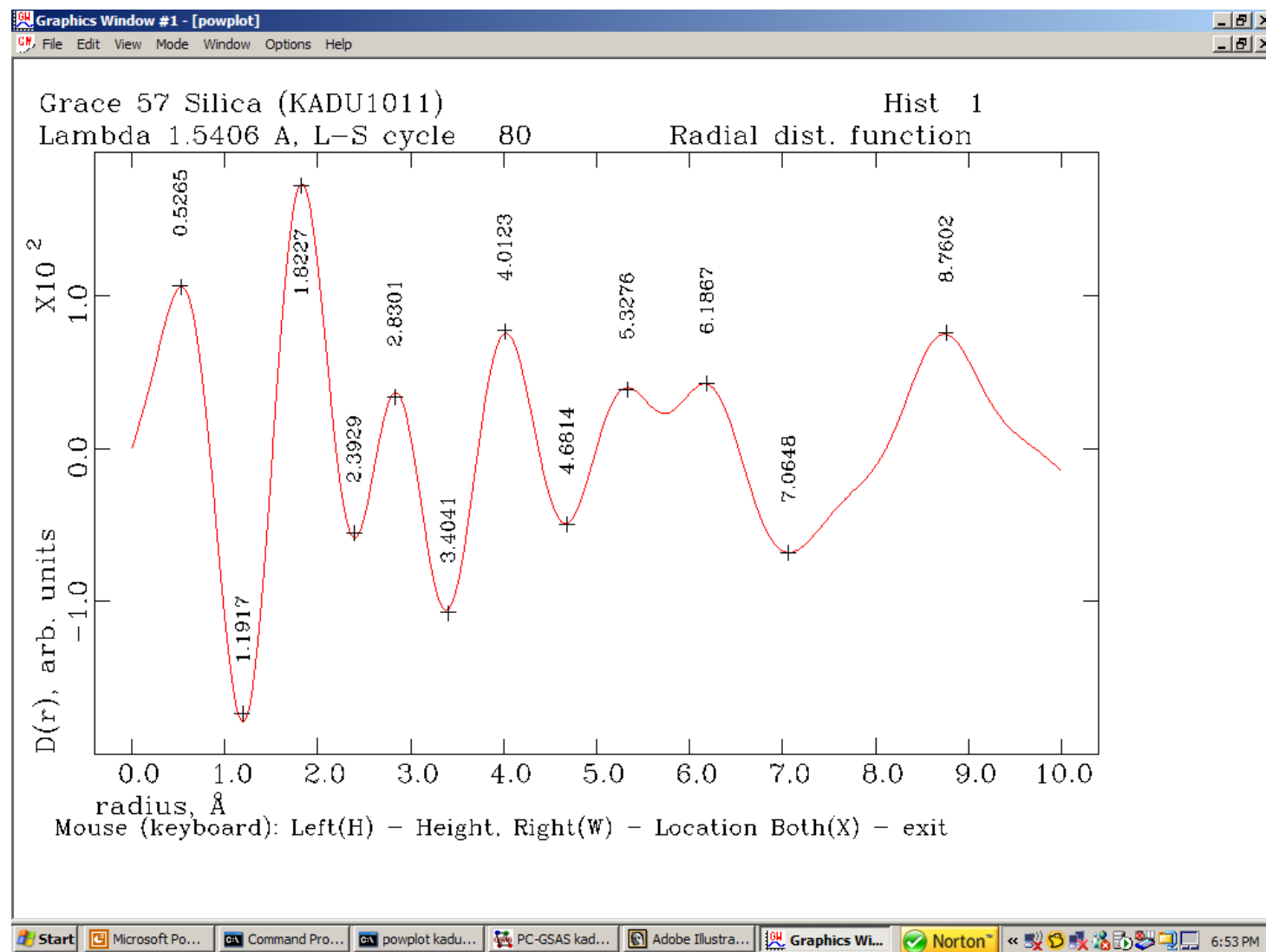
The refined model yields a real-space radial distribution function



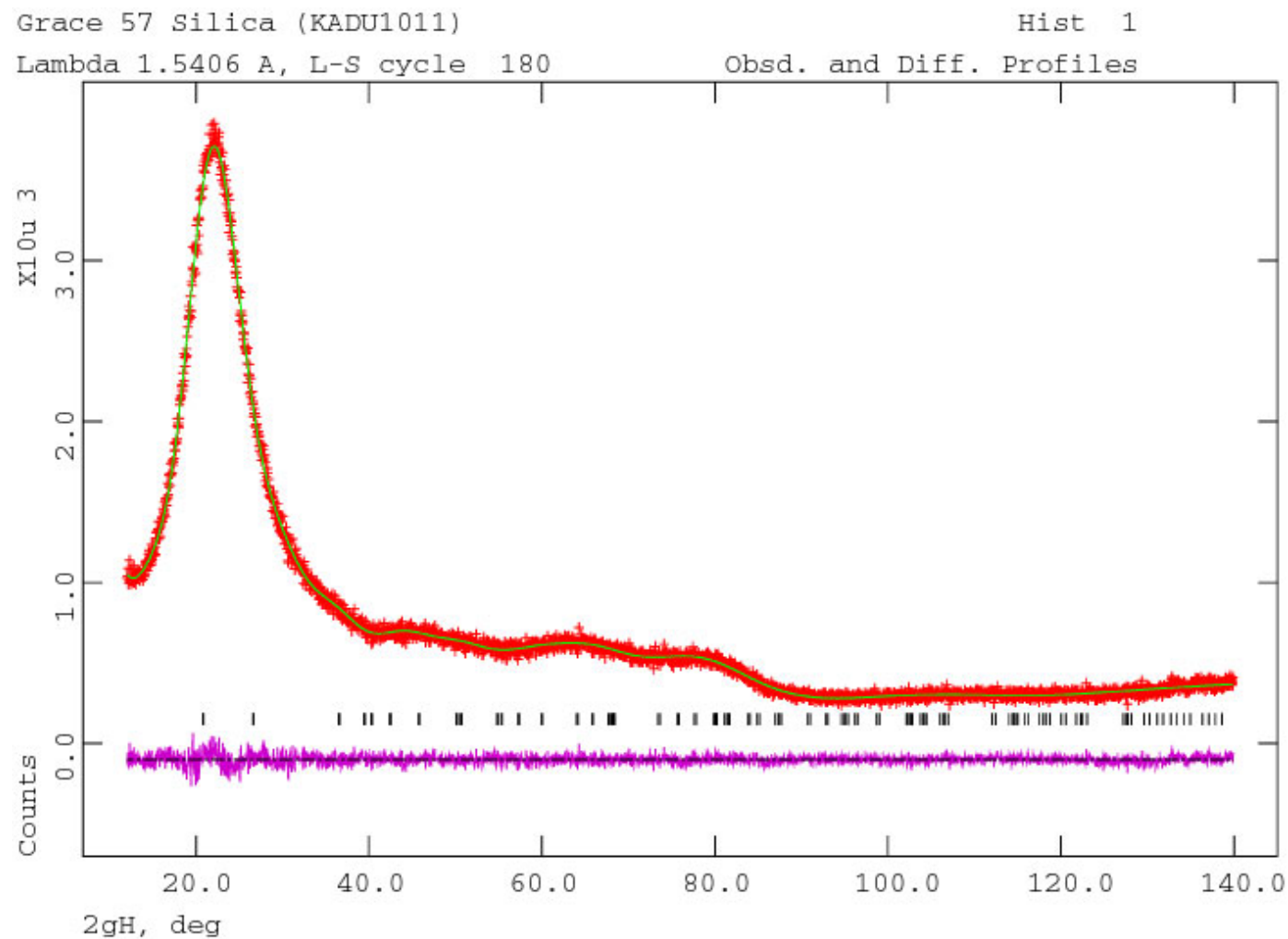
Iterating the Debye model steadily improves the diffuse fit



RDF peaks index directly to coordination shells



The final diffuse fit converges with chi-squared = 1.11



Lab XRD alone cannot settle apatite site occupancies

- $\text{Gd}_{8+x}\text{Ca}_{2+y}(\text{SiO}_4)_6\text{O}_{2+3x/2+y}$ system with anion/cation vacancies
- **Lab XRD is insufficient to answer:**
 - Impurity wt%?
 - Volume change?
 - Cation site preference?
 - Meta-prism twist?

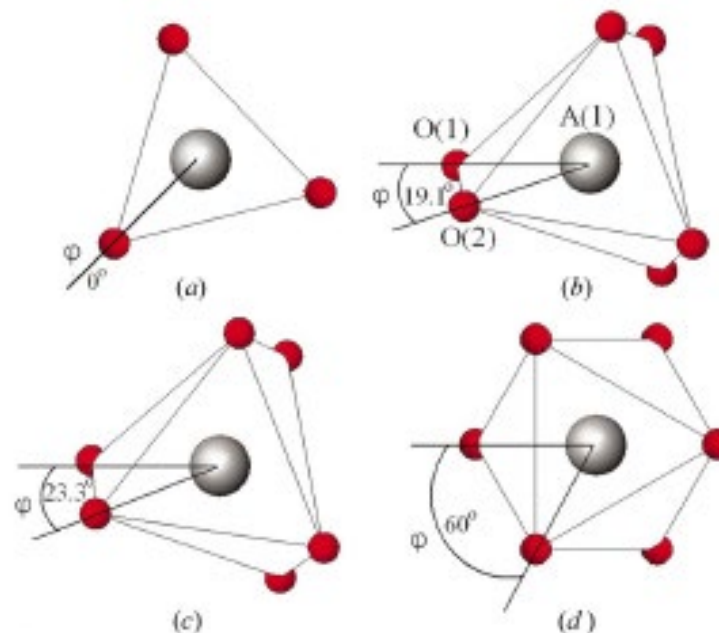
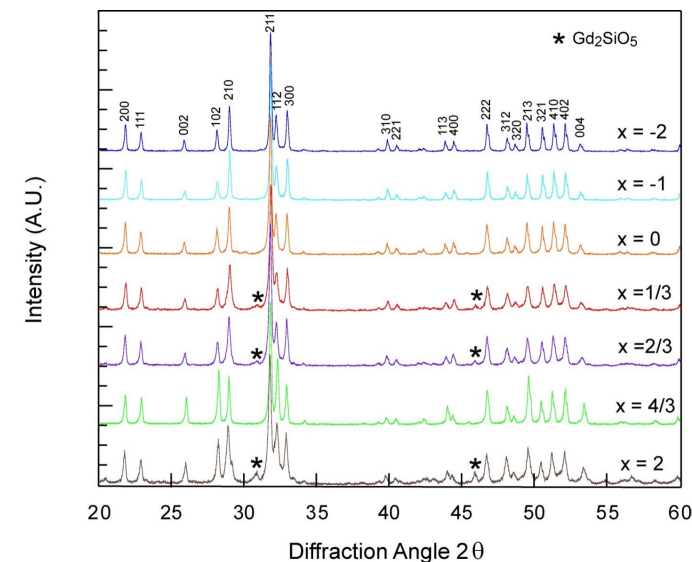
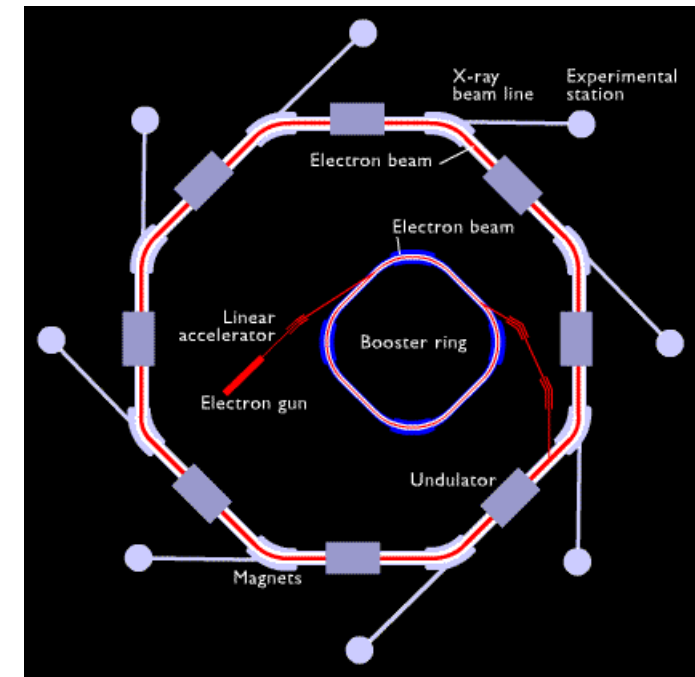
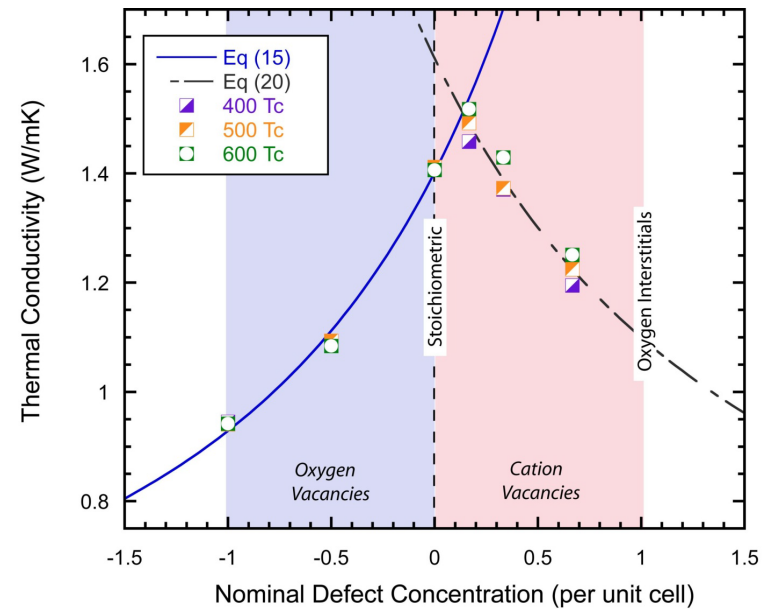


Figure 6
Twist angles of $A(1)\text{O}_6$ polyhedra in (a) models I and II, (b) chlorapatite, (c) fluorapatite, and (d) model III (as found in glaserite).

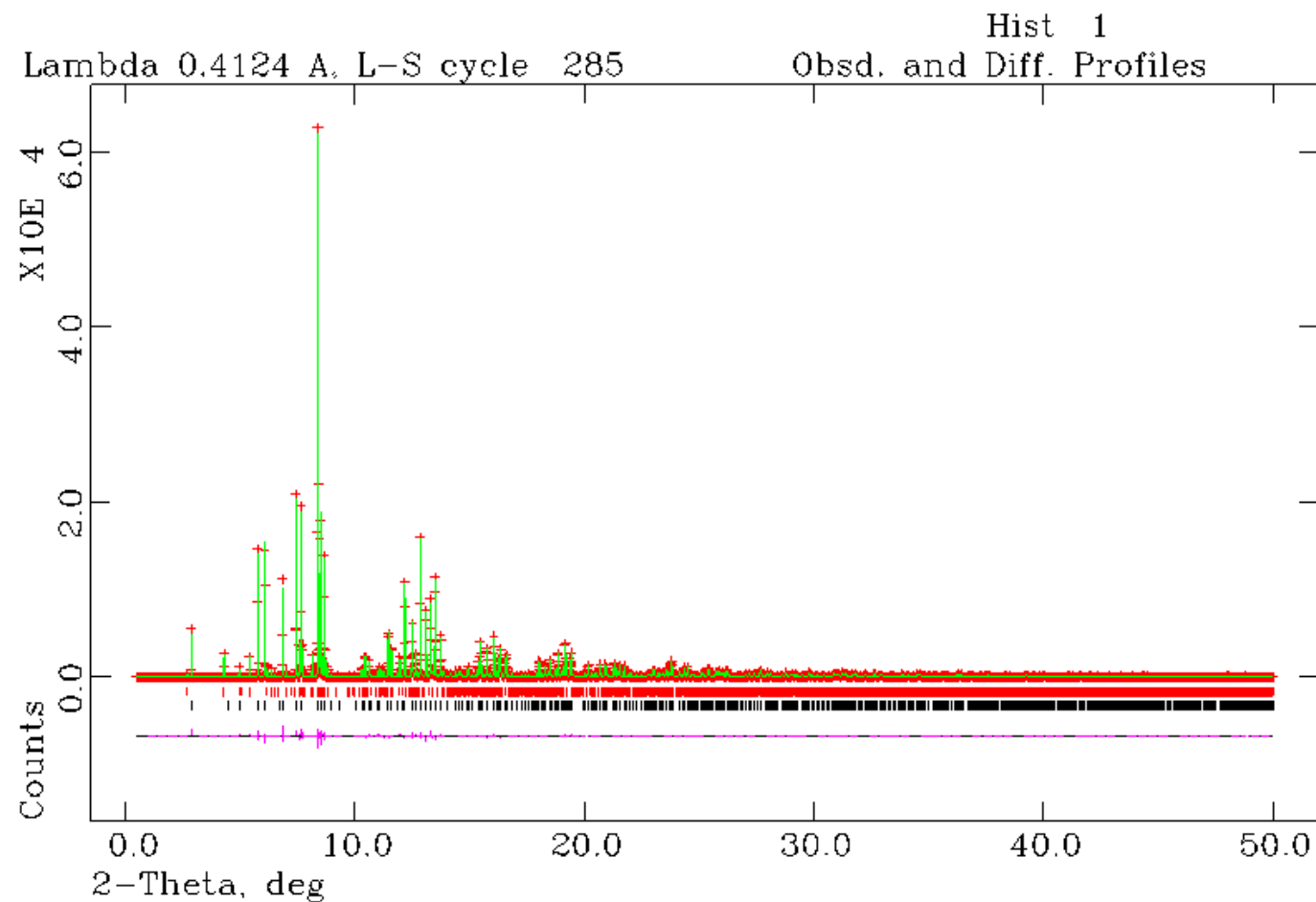


Synchrotron data adds the resolution occupancy refinement needs

- Stoichiometric composition should have the highest thermal conductivity
 - Phase separation?
- **Synchrotron diffraction brings:**
 - Improved signal-to-noise ratio
 - Improved resolution
 - Minimized anomalous scattering



Refinement finds 5% Ca on A1 and a 23.6 degree meta-prism twist



Rwp = 6.29%

85% $\text{Gd}_{7.7}\text{Ca}_{2.3}(\text{SiO}_4)_6\text{O}_2$

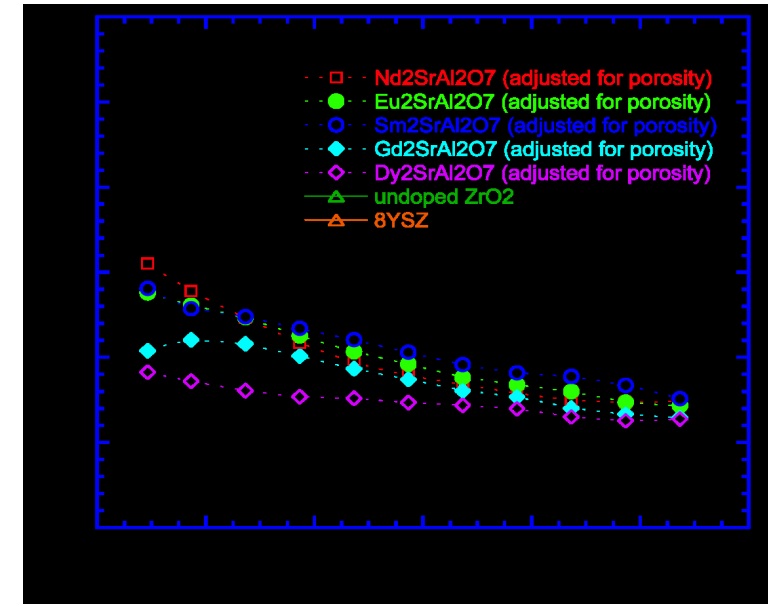
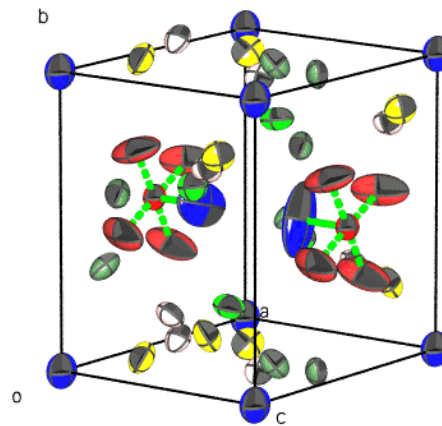
14% Gd_2SiO_5

Ca has ~5% on A1

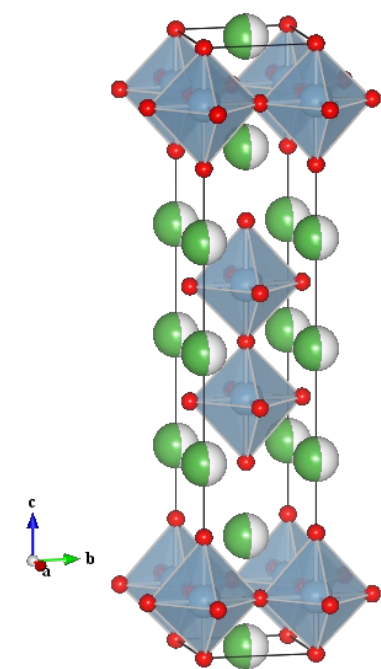
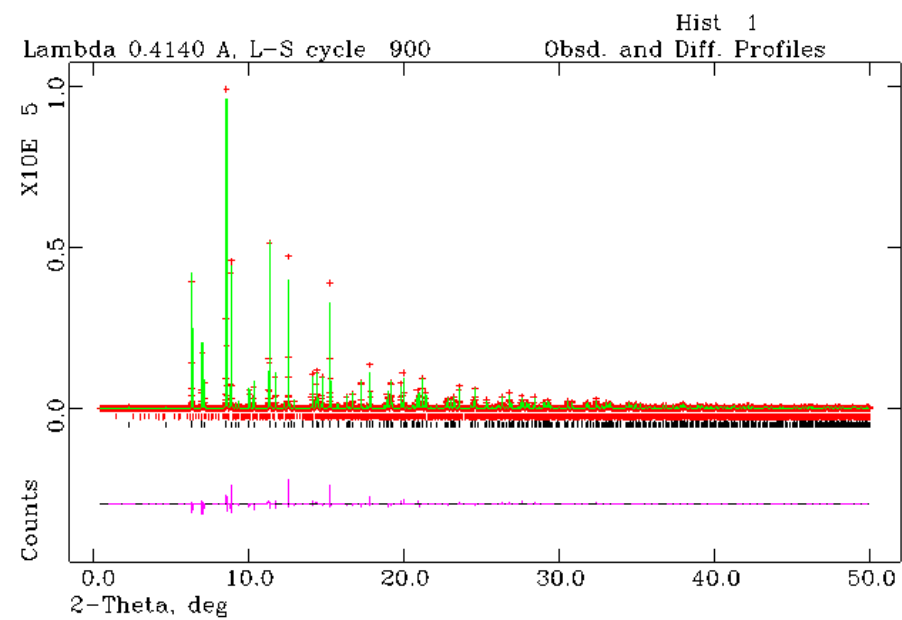
Meta-prism twist 23.6°

Large refined Uiso values can flag rattling atoms

- Thermal vibrations are quantitatively studied in Rietveld refinement
- Uiso, or Uij for anisotropic vibration
- **Large U values could be due to rattling**
- An explanation of RE₂SrAl₂O₇ thermal conductivity?

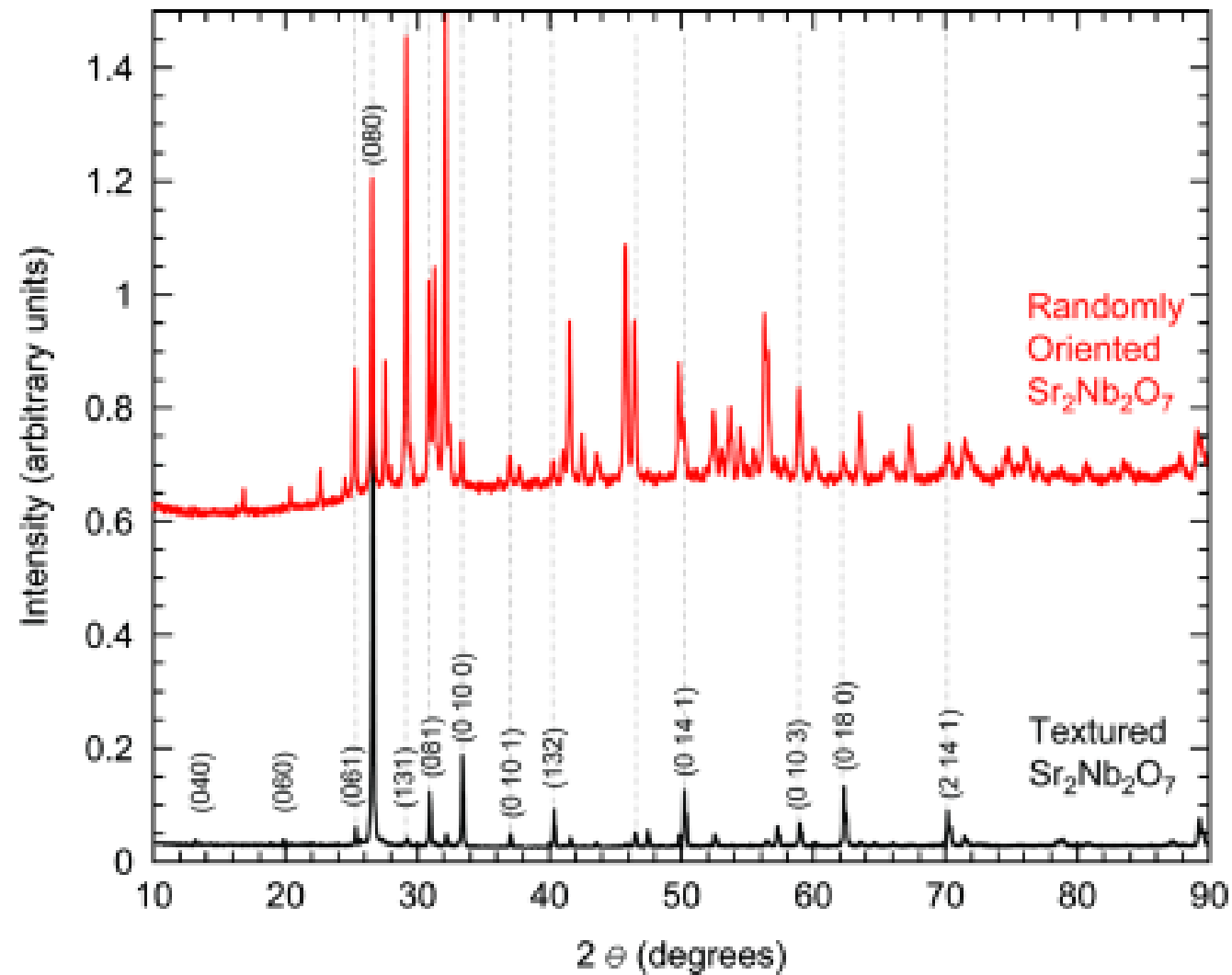


Occupancy and Uiso trends track across the RE2SrAl2O7 series



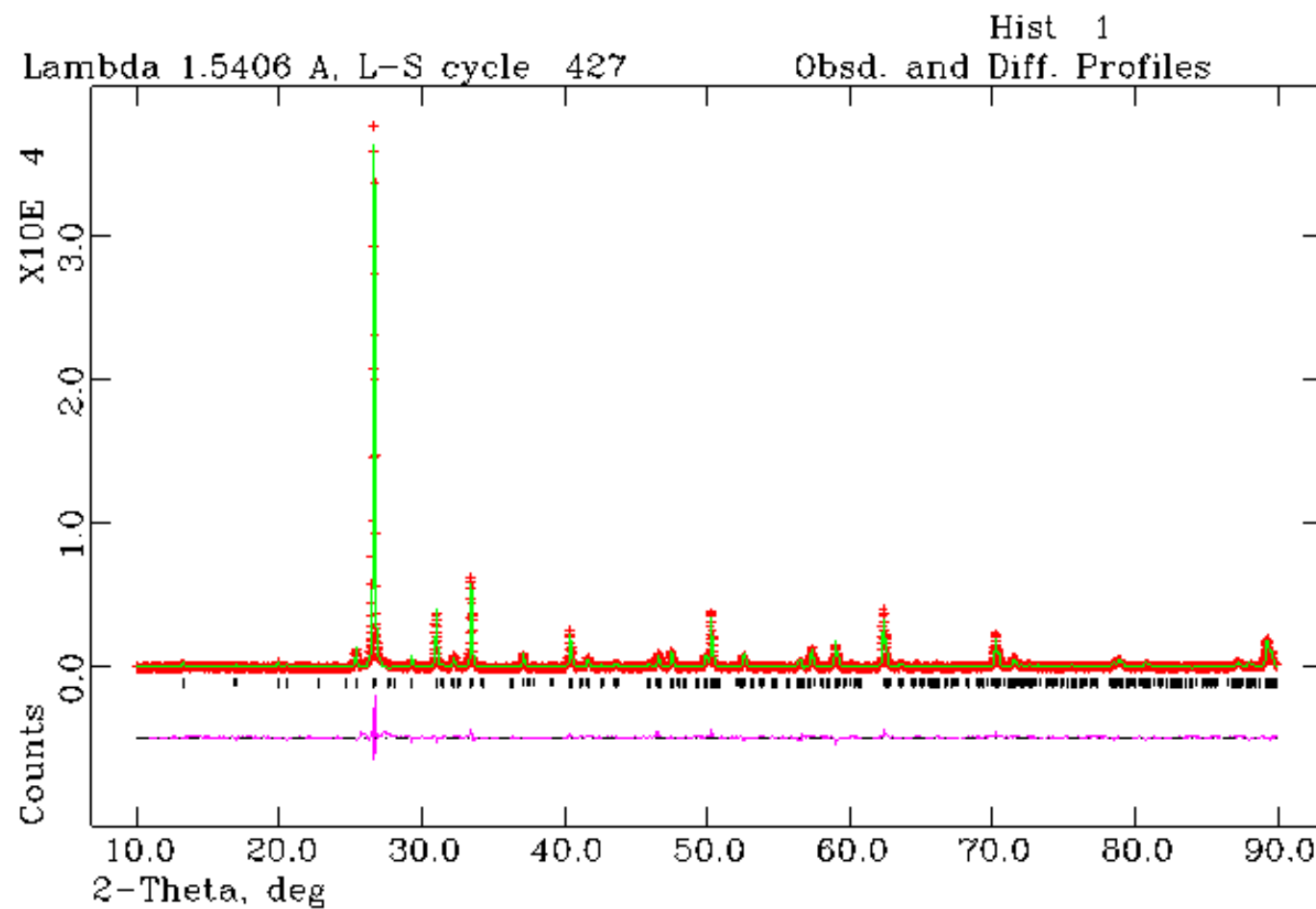
Composition	La2SrAl2O7	Nd2SrAl2O7	Sm2SrAl2O7	Eu2SrAl2O7	Gd2SrAl2O7	Dy2SrAl2O7
Uiso A1	0.11(2)	0.21(5)	0.40(3)	0.20(1)	0.37(3)	0.96(2)
A1 (P2) occupancy	.73	.54	.43	.32	.25	.21
A2 (RS) occupancy	.27	.46	.56	.67	.72	.79

Texture is obvious by eye; Rietveld puts a number on it



$\text{Sr}_2\text{Nb}_2\text{O}_7$: clearly textured to the b-axis, but what is the preferred orientation factor?

March-Dollase refinement gives Sr₂Nb₂O₇ a texture index of 5.52



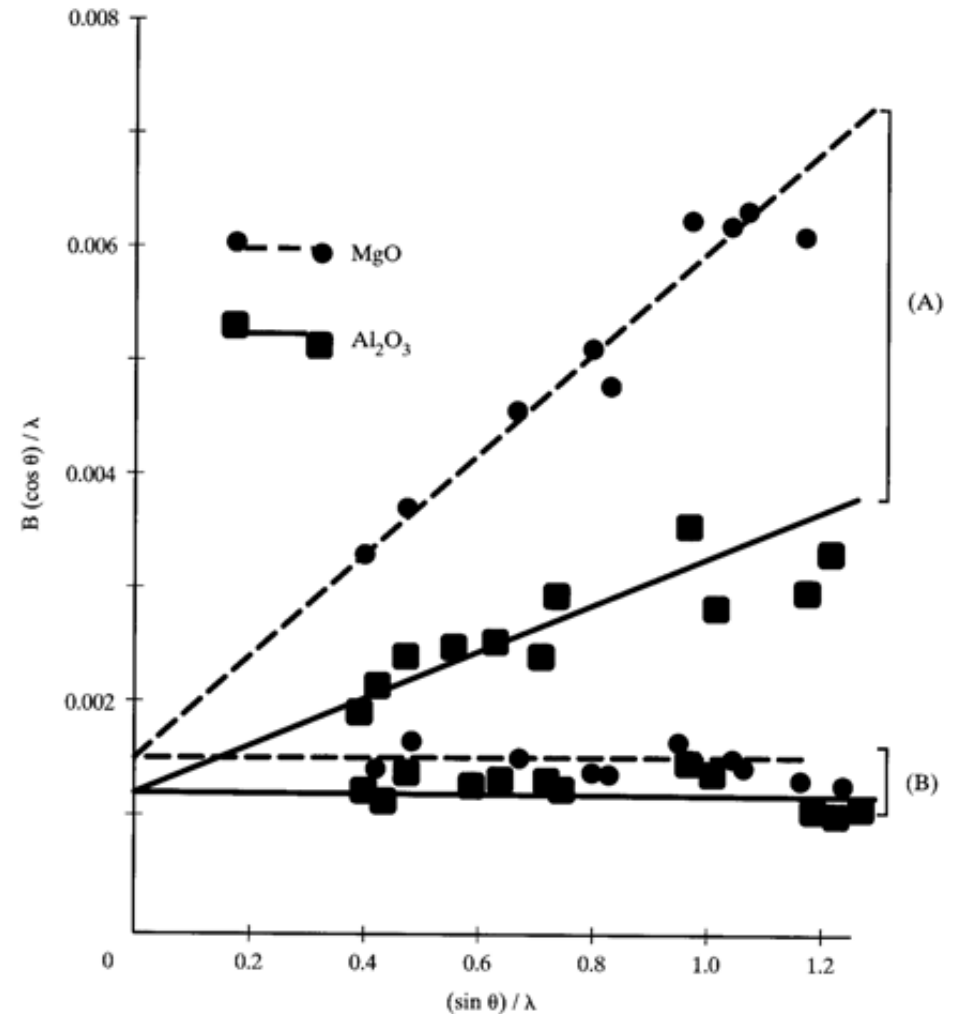
Rwp = 6.9%

Texture index = 5.52

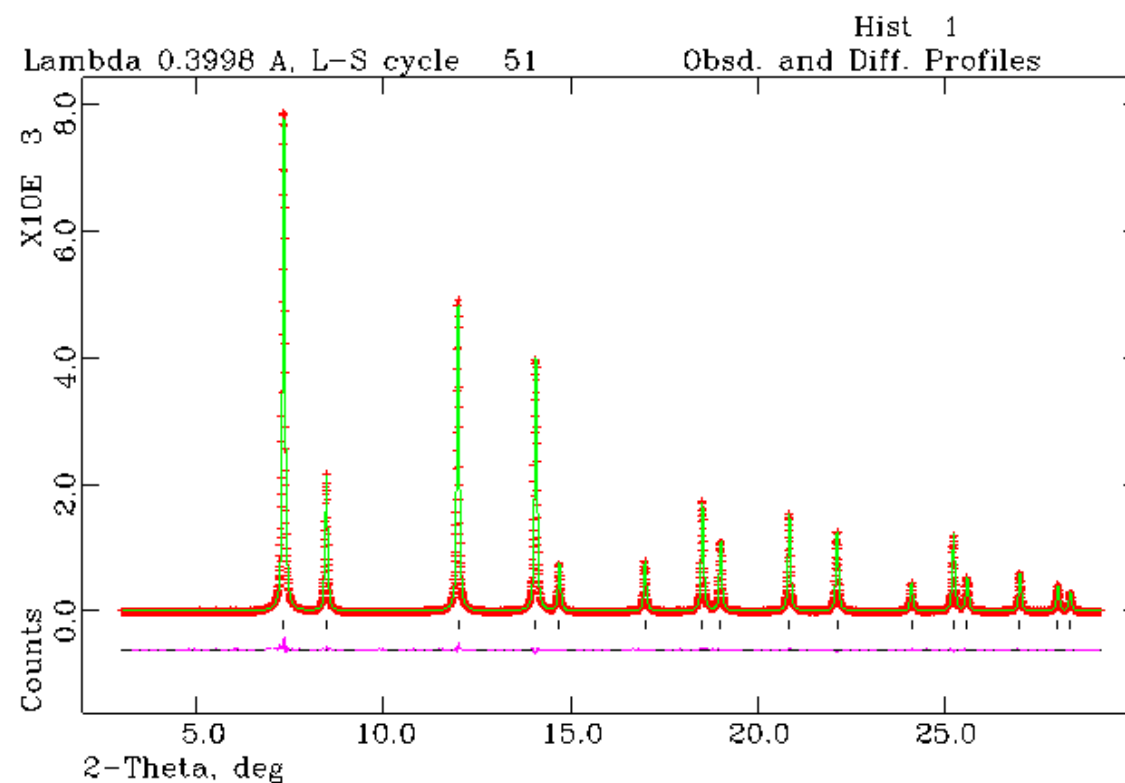
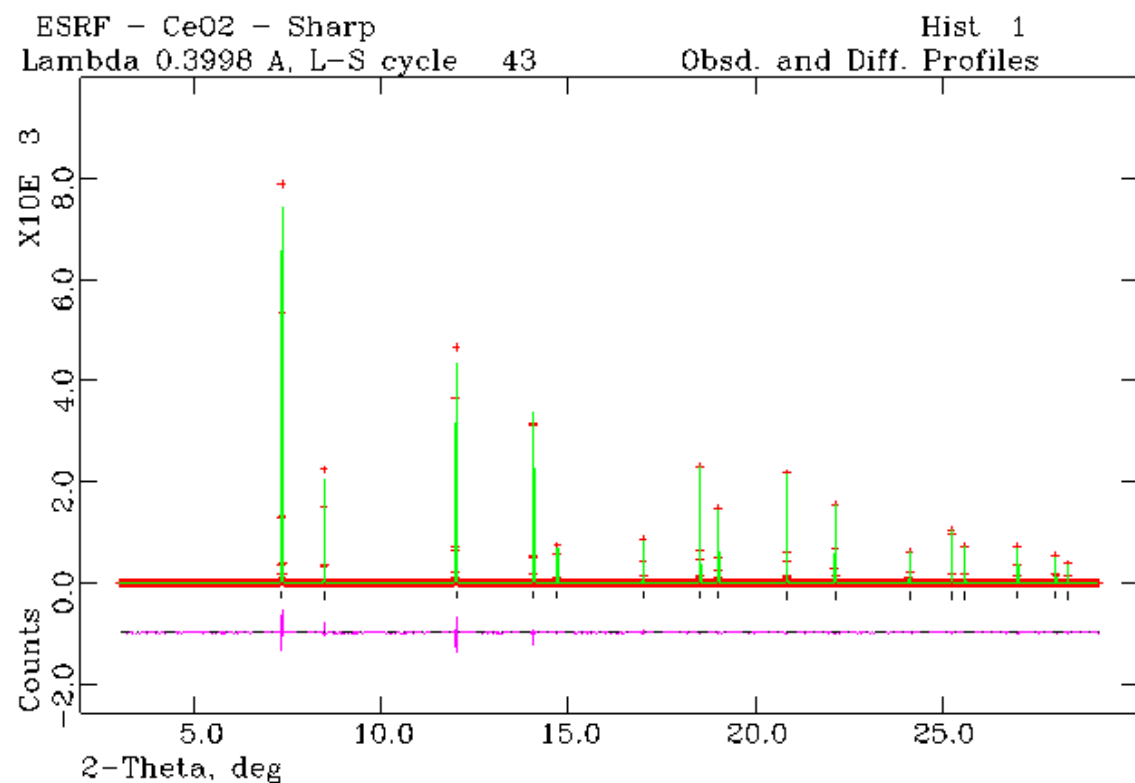
Note: the actual compound had a small amount of La³⁺ doped on the Sr²⁺ site

Calibrate the instrument first; leftover broadening is your sample

- Characterize instrument shape parameters using a standard
- Then U, V, W give the integral breadth for Scherrer and Stokes analysis
 - $Dv = K\lambda / (\beta \cos\theta)$
 - $\epsilon = \beta / (4 \tan\theta)$
 - $(\beta_{\text{obs}} - \beta_{\text{inst}}) \cos\theta = \lambda/Dv + 4\epsilon \sin\theta$
- **Williamson-Hall:** plot $\beta \cos\theta / \lambda$ vs $4\sin\theta$; slope and intercept give strain and size



Micronized CeO₂ refines to 226 Angstrom domains and 0.011% strain



Rietveld refinement is a complete quantitative toolkit

- **Crystal structure information**

- Atomic positions, thermal displacement parameters, occupancy

- **Beyond the average structure**

- Preferred orientation, size, strain, quantitative phase analysis (including amorphous material)

- **And remember**

- Judge fits with the profile plot and chemical sense, not Rwp alone
- Turn parameters on in a correlation-aware order

